

Medicina Gymnastica:
O R, A
T R E A T I S E
Concerning the
P O W E R of E X E R C I S E,
With Respect to the
Animal Oeconomy;
A N D T H E
Great Necessity of it in the Cure
O F
Several Distempers.

By *FRANCIS FULLER*, M. A.

The Second Edition, with Additions.

L O N D O N:

Printed for ROBERT KNAPLOCK, at the
Angel and Crown in St. Paul's Church-Yard.
MDCCV.

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Title: Medicina Gymnastica

Author: Francis Fuller

Release date: October 18, 2020 [eBook #63497]

Most recently updated: October 18, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/63497

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*** START OF THE PROJECT GUTENBERG EBOOK MEDICINA
GYMNASTICA ***

We whose Names are here Subscrib'd, having perus'd this Discourse, Entitled, *Medicina Gymnastica*, &c. Judge it well Worthy to be Communicated to the Publick.

*Datum in Comitiis Censoriis
ex ædibus nostri Collegii,
Febr, 2. 1704/5.*

Edwardus Brown, Præses.

*Walt. Charlton,
W. Vaughan, } Censores.
Walt. Harris,
Jo. Hawys,*

Joh. Bateman, Regist.

Medicina Gymnastica:

OR, A

TREATISE

Concerning the

POWER of EXERCISE,

With Respect to the

Animal Oeconomy;

AND THE

Great Necessity of it in the Cure

OF

Several Distempers.

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THE PREFACE.

The Generality of Men, have for a long time had too Narrow Thoughts of Physick, as if it were in a manner Confin'd to little more than Internals, without allowing themselves the Liberty of common Reasoning, by which they easily might have found that the Humane Body is liable to, and requires several Administrations of a very Different Nature, and that it is very unreasonable to suppose, that since there are so many ways for Diseases to enter upon us, there should be so few for Health to return by. Internals do indeed make up the far greatest part of the Means of Cure, but yet there are Considerable Cases, where the very Nature of the thing requires other Methods; and this would appear very obvious, if it were not for our too Partial Consideration of the Body of Man, by attributing too much to the Fluids, and too little to the Solids, both which, tho' they have a Mutual Dependence upon one another, yet have each of 'em some Proprieties, and if out of Order, require something particular in the Application to restore 'em again. Consent in the Solids, answers to Mixture in the Fluids; and as an ill Ferment, as soon as it comes into the Blood, diffuses and mixes it self with the whole Mass, and cannot often be extirpated, till the Medicine given for that end, has been taken so long as to be diffus'd and intimately mix'd with the Blood likewise; so a Violent Seizure in one of the Solid Parts, commonly draws all the rest into Consent, and a particular Application to the Place primarily affected shall do no good, when a Universal one shall Cure; and a thing which would be trivial and Vain, if us'd as a Topick, shall by a Universal Administration prove of the greatest Importance. We see Contraries often prove Remedies to one another in the Juices, and Poisons become Beneficial, when oppos'd to certain Humours, why should we not then allow of the same Rule, in the Containing Parts of the Body? If by a Supine Course of Life, the Nervous parts are weakned and relax'd, why should we not suppose the contrary way of Living, the most likely to repair 'em? Since the Vigour of those parts is acquir'd by Use; they are the Active part of the Man, and not always liable to the Impressions of the Fluids, for tho' you invigorate the Blood

ever so much by the most generous Medicines, the Nerves may remain Effete and Languid notwithstanding; but if the Nervous parts are extended and exercis'd, the Blood and the Humours must necessarily partake of the Benefit, and soon discover it by the Increase of their Heat and Motion. There is so much of a Relative Nature in every thing, that can concern the Health or Distemperature of the Individual, that there is scarce any thing so prejudicial, or seemingly Absurd, but may in different Circumstances prove as Advantageous. The World has lately had full Evidence of the good Effects of an Intense Cold, equally apply'd to all the Parts of the Body at once, which Method of Curing would, not many Years agoe, have been thought very Extravagant, and certainly Destructive. On the contrary, there are other Cases, where a Warm Bath is only prevalent; and though some People have suppos'd it to be only a kind of a last Resort, when other things have been try'd in Vain, yet it is quite otherwise, it being impossible to remove some Diseases of the Limbs, without an universal equal Relaxation. Again, quite different from this is the equal Distribution of a greater Degree of Heat throughout the whole Body, which is procur'd by Habitual Exercise; in the former Method, the Parts are relax'd, in this they are strengthened, and in every Respect the Effects are widely different, tho' in both ways there is a considerable Encrease of Heat. But to carry this enquiry farther, there are some Distempers, and those not altogether so rare neither, in the Cure of which no Positive Physick of any sort whatsoever, can be serviceable, nothing but a gradual Substraction of the Cause, an Alterative abstinence, if I may so speak, being necessary; as there have been some Gouts in some temperate Persons, of a strong and rank Constitution, which nothing could remove but a very low Diet, and an entire Abstinence from Flesh; to them Flesh being as Wine is to others, who Contract that Distemper by their Excess: and the same Observation holds good likewise in the Ophthalmia, and some other Cases of the Eyes, as any that will duly weigh the Circumstances of some Persons often subject to 'em, will find Reason to believe, so that some Men are by their Constitutions condemn'd to an Antidiluvian Diet of Roots and Vegetables, or else to suffer worse Inconveniencies; and when those happen, it is absurd to expect, by the most Celebrated Remedy to cure the Disease, when the Cause continues. These are Instances of several Methods, widely differing from one another, and yet of absolute Necessity in their particular Cases, which shews us, that we ought not to be so eager after Courses of Pharmacy in all Cases, without

distinguishing where other ways are most rational. It is one thing to dispose Nature to collect her own Strength, and throw off her Enemy; and it is another to assist her by the Corpuscula, the Minute parts of a Medicine given inwardly; the first way has Regard to the whole Animal Oeconomy; the second reflects the Blood and Juices chiefly; the first may succeed, where the second cannot, because here the Laws of Motion, and the Rules of the Oeconomy are enforc'd, and brought to be assisting to a Recovery of Health, which in some few Cases can't be effected by a private and simple Attempt upon the Blood only.

These are Notices which don't lie so far out of the reach of any Man, that will give himself leave to animadvert upon the Misfortunes of his Neighbours, but that they may discern'd; and if they were duly considered, there could be no room for Empiricism, for that is founded on the Folly, or Negligence of Mankind; and if Men will give themselves up to Expectation, and resolve to be always alike Passive, in Hopes the Miracle will be wrought upon 'em, without any Concurrence on their part, it is but just they should suffer something from the Pedantry, the little things of the Profession, which they are so dispos'd to receive.

As for the Exercise of the Body, which is the Subject of this ensuing Discourse, if People would not think so superficially of it, if they would but abstract the Benefit got by it, from the Means by which it is got, they would set a great Value upon it; if some of the Advantages accruing from Exercise, were to be procur'd by any one Medicine, nothing in the World would be in more Esteem, than that Medicine would be; but as those Advantages are to be obtain'd another way, and by taking some Pains, Mens Heads are turn'd to overlook and slight 'em. The habitual increasing of the Natural Heat of the Body, as I took Notice above, is not to be despis'd; but if we consider that it is done without charging Nature with any subsequent Load, it ought to be more valuable, for I may by some generous Medicine, or a Glass of Wine, raise Nature to a great Pitch for a time, but then when these Ingredients come to be digested and resolv'd into their Principles, Nature may be more oppress'd with the Remains of the Medicine, than she was at first reliev'd by it: Therefore if any Drug could cause such an Effect, as the Motion of the Bodies does, in this Respect, it would be of singular Use, in some tender Cases upon this very Account; but then add to this the great Strength, which the Muscular and Nervous parts acquire by Exercises, if

that could be adequately obtain'd likewise by the same Internal Means, what a Value, what an extravagant Esteem, would Mankind have for that Remedy, which could produce such wonderful Effects? But since those Benefits are to be procur'd another way, how difficult is it to bring People to conceive it? To build up such a Belief in the Minds of Men, is to raise a Structure, the Foundations of which, can be laid with no less Difficulty, than the removing of the Rubbish of a Vulgar Error.

As for what I have said, relating to the Balsamick Method in the Cure of Consumptions, tho' I may perhaps give Offence to many, yet I am sure I speak the Thoughts of some of the greatest of the Profession, if not of the generality of Physicians, who have for a great while, had a secret Regret, at the ill Success of those Means, which nothing could have stifl'd and suppress'd, but the Difficulty of agreeing to substitute better. This I do not pretend to do neither, this would be too daring an Enterprize for me; I only point out some few Mistakes, which one of my little Consideration, may sometimes discern in the Actions of those of greater Condition. And perhaps when these Mistakes are rectified, the Business is done, as well as if I had discovered to the World, some wondrous Medicine, dignifi'd with as many and as swelling Titles, as an Eastern Monarch; for it seems to me to be a great Mistake, to wait for some Medicine of a great Character, to be the Specifick, or Certain or Adequate Remedy of this Disease. I am inclin'd to think, that the Cure must result as much from some Circumstances of Management, as from any Medicine. I believe we have the best of Medicines for this Case, in our Hands, if we did but take Pains to obviate some minute Circumstances, that make those Medicines often prove deficient; if the Success did not vary by Reason of such small Emergencies, I don't know how we should account for what we find to be true, viz. that some common and despicable Medicines shall in one Person produce a wonderful Cure, and in another do nothing at all, tho' as far as Humane Wisdom can discern, the Case is exactly the same in both; and why should it seem strange, if this should happen in Cases, so fine as those of the Lungs, when we see in several Manufactures the Accuracy and Perfection of an Operation shall depend upon a certain Finesse, the Criterion of which is scarce to be describ'd, but to be learn'd only by Numerous Repetitions. A Man may be told the Ingredients of a Composition, and the way of working 'em up; may see it done, and not be able to distinguish the Nicety of the Circumstances, which conspire to make the Work consummate. Do not we

know, that in the Dyers Trade, (to give but one Instance) their Mixtures shall sometimes miscarry, and the best Artist not be able to shew a Reason for it; now if it is thus in matters of Artifice, I think we may with great reason suspect something of a like Nature, in some few Cases of the Humane Body; in these Maladies of the Breath, there is a Finesse peculiarly distinguish'd from the Circumstances of any other Diseases, as is apparent from the Power of so fine a Body as the Air; and therefore we need not think it so strange, that a little matter should be able to give a good or bad turn. And if this will be allow'd to be good Reasoning; what can be so likely to obviate such Miscarriages, as a due application to that most moderate and easie Exercise, which I have apply'd to the Cure of this Distemper? That a mild Exercise will do something like this, may be learnt from the Effects of those which are more violent; we know very well what has follow'd upon going into the Bagnio, after violent Exercises, the Blood of those Persons who have done so, has so been exalted in its Crasis, as to be upon that very account, fatally Preternatural; tho' upon the first Thought, One would be apt to conclude, that such violent Perspirations should rather impoverish the Blood; but it is quite contrary, just as in Hot Climates, tho' People Sweat profusely, yet they are rather the more Brisk and Lively for it. And thus Moderate Exercise, by Augmenting the Natural Heat of the Body, will enrich the Fluids, and by encreasing the Velocity of the Circulation, every the minutest Particle will be brought much oftner to the Test of the Strainers, than otherwise it would have been; so that both the Venous Fluid and the Spirits will after an Eminent Manner be exalted, and as it were Rectifi'd in the making. Therefore it cannot be a doubt, that these means may improve a Medicine, when it comes into the Blood, supply its deficiency, and remove the Inscrutable Inpediments in the Cure of so nice a Disease, as this I am speaking of.

These things are not owing to Notion and Speculation, but to Practical Observation; I cannot tell otherwise how to explain what I have seen to be the Effects of such means; I have seen a Poor Gentleman at Hampstead, in the most deplorable extremity of a Consumption, betake himself to Riding twice or thrice a day, upon his Waters, put a stop for a time to his Distemper, and keep Death as it were at a Bay for some Weeks; which plainly shew'd, that if he had done so sooner, his former Medicines would undoubtedly have took effect; and how shall I account for this, but after some such manner as I have here insisted on? This is the Sum of my

Thoughts on this Case, and if it does not strike the Imaginations of some People with so much Surprise, as other Pharmaceutick Discourses, I cannot help it; nay, if any should be so hard upon me, as to alledge that I am quite mistaken in the Cure of this Disease, when far advanc'd, let it be so; yet I hope they will allow me thus much, that it appears plain enough, for the Nature and Power of Exercise, that we ought to distinguish in this Distemper in like manner, as we do in another, which it would not be good Manners to mention, at the same time with the Consumption of the Lungs; in which latter Case, if a Young Debauchee happens to get a Misfortune, he does not let it run on till it come under another Denomination, but looks out for a present Cure, which may then be certainly obtain'd; and so I think it is no less obvious, if Exercises be as efficacious as I assert, that it is our Choice, whether a Cough shall run on to a Consumption; that is, I understand, that when a Cough comes to last above a Month, and begins to chop in its Indications, to require one while incrassating Medicines, another while attenuating ones; I say it is high time to look out, and set upon a resolute Course of Riding; that by a Series of Journeys, continued Day after Day without intermission, the obstructed Perspirations may be restor'd, which may then be easily effected, because the Body not being then much impair'd, so many Millions of Succussions coming close upon one another, must affect both the Strainers of the whole Body, and the Juices to be strain'd; and he who can oppose a Truth so manifest, must fly in the Face of Nature, and deny the Principles of the Oeconomy.

Having had some Occasion to mention the Ancients, and since Gymnastick Courses made a great part of their Physick, perhaps some will think I am setting up for one of those who cry up the Ancients upon all occasions, out of pure Enmity to the Authors of Modern Discoveries, but they will find themselves mistaken; I neither affect to be of a Party in Physick, or Singular; I seldom yet knew any one, who cry'd up Acids, or cry'd down Bleeding, who extoll'd the Ancients universally, and vilifi'd the Moderns, but if he really understood what he pretended to, had some By-end to serve in so doing. I owe the Ancients no more Respect, than what is due to those upon whose Shoulders we stand, and upon whose Rudiments we have Built; we know their Ætiology was all wrong, their Pharmacy in general was Rough and Barbarous. (I say in General, for there are some few Cases in which their Medicine will perhaps last for ever,) but yet the ways they took to supply their want of Medicine were very commendable,

and may shame us; their exquisite diligence in establishing certain Diagnosticks, and their Recourse to various Exercises, to help out what they wanted in other means; 'tis the last of these which has given me occasion to take Notice of 'em, and I think it will appear in the following Discourse to be of so much Importance, that not only in their days, but in ours also, the Art of Curing may be said to be in some Measure imperfect without it, there being some Cases in which the Use of Exercises will be absolutely necessary as long as Nature shall last.

I have this more to add, that I did not take this Subject out of Choice but Necessity, 'tis not owing to Theory and Speculation, but Experience; the severest Experience, which my own Misfortunes have given me but too much occasion to make in a Distemper, which some Years ago I happen'd to be afflicted with, as I have related in the Appendix to this Treatise; 'twas under that severe Discipline I made most of the Observations of this Treatise; and tho' mine was an Anomalous and singular Case, yet from what I perceiv'd to be the immediate Effects of that Exercise in common to all, and from the manner by which it caus'd my Distemper to give way, I could not but discern in what other Cases it was likely to be of the same Consequence, and I have had so very many Serious and Calm hours to confirm me in these Notions, that I cannot think I have extended 'em too far.

As for what Reception these Papers are like to meet with, tho' I have ventur'd abroad in a Cause so obsolete, in an Age so fruitful in Pharmacy, and abounding in Splendid Discoveries; and tho' I am destitute of a New Hypothesis, that Specious Image of Truth, that Idol to which the Learned all bow down; yet, if what I have advanc'd be strictly Conformable to Truth, and of real Necessity in some few Extremities, I hope I may pass, upon the Merits of my Subject, tho' without Flourish and Ornament.

Medicina Gymnastica:
OR, A
TREATISE
Concerning the
Power of Exercise
With Respect to the
Animal Oeconomy.

That the Use of Exercise does conduce very much to the Preservation of Health, that it promotes the Digestions, raises the Spirits, refreshes the Mind, and that it strengthens and relieves the whole Man, is scarce disputed by any; but that it should prove Curative in some particular Distempers, and that too when scarce any thing else will prevail, seems to obtain little Credit with most People, who tho' they will give a Physician the hearing when he recommends the frequent Use of Riding, or any other sort of Exercise; yet at the bottom look upon it as a forlorn Method, and the Effect rather of his Inability to relieve 'em, than of his Belief that there is any great matter in what he advises: Thus by a negligent Diffidence they deceive themselves, and let slip the Golden Opportunities of recovering, by a diligent Struggle, what could not be procur'd by the Use of Medicine alone.

Whether this proceeds from the Customs of these Northern Nations, so different from those of the Ancients, and of more Southern Countries, who seem to have plac'd almost as much in their Methods of Exercise, as in their internal Physick; or whether from the narrow Notions most People have conceiv'd of the Art of Physick, as if it imply'd little more than Internals only, without considering that External, Mechanical, and all other Means whatsoever, that give Relief, properly belong to it; this I shall not pretend to determine, but this I think I may venture to affirm, That most Men indulge

themselves in the Expectation of more sudden Relief, than the Nature of the Case will admit of, as if they thought that Medicine was always to take like a Charm, without putting 'em to the Expence of much Time or Pains; they do not consider the wonderful Variety of the Disorders of Nature, and the Stubbornness of some Cases, which will not permit the Sick to be wholly passive, but indispensably oblige him to conspire with his Physician, and strive indefatigably to exalt his Constitution to a degree requisite to supply the Defect of internal Physick; which industrious striving on the part of the Sick, being what is here meant by Exercise, and which it is my purpose to represent, as more efficacious than it is generally believ'd to be; I think it proper, first to explain what I mean by it in this place.

By *Exercise* then, I understand all that Motion or Agitation of the Body, of what kind soever, whether voluntary or involuntary, and all Methods whatsoever, which without the Use of Internals, may (or without which Internals alone may not always) suffice to enable Nature to expel the Enemy which oppresses her; confining my self to the Consideration of it only as it may prove Curative, not as Palliative, or barely Preservative.

And here, before I attempt to demonstrate how Exercise proves so beneficial in some few Cases, it may not be amiss to premise briefly some of the ways Nature takes to relieve her self, when in danger of being oppress'd, which may serve to illustrate my following Discourse; as likewise some Instances of the Efforts of Nature, caus'd by external Application, or at least by such Internals as cannot be suppos'd to be Cordial or Vinous, or to be assimilated with the Blood and Animal Spirits.

First, then; We may observe how Nature acquits her self of what we commonly call a Cold, wherein a considerable quantity of the *Materia perspirabilis* is detain'd, by a sudden Constriction of the Pores of the Skin, we shall find, that after some time the saline Particles growing turgid, vellicate some Fibres of the fifth Pair of Nerves dispers'd about the Nose, which by consent draw the Diaphragm into a Convulsive Motion, by which the Air is press'd out of the Lungs thro' the Nose with some Violence; and by the shock the Glands of those Parts are open'd, and the Humour, which was detain'd, is let out. This is *Sneezing*; to which frequently is join'd *Gauiping* or *Retching*, another Method of Nature to shake off a Load that she finds growing upon her; this is more often repeated than *Sneezing*, and may be conceiv'd to dilate some internal Parts by those Stretchings out of

the Limbs, and more plentiful Inspiration; *Sanctorius* tells us in the 31st *Aphorism* of his *fourth* Section, that, *Corpora Oscitatione & Pandiculatione horæ dimidiæ spatio magis perspirant, quàm tribus horis alterius temporis*, and compares it to a Cock's clapping his Wings after his Rest; so that it is manifest, that even in that vulgar Affect there is an Alteration caused in the Body, that is not Contemptible.

To these may be added the *Singultus* or Hiccough, by which the Ventricle, when too full, endeavours to relieve her self, either by throwing off some of the rarifi'd Contents by the Gullet, or perhaps by the *Pylorus*.

Lastly, Let us consider how *Laughter* affects us, and it will appear, that this Contrivance of Nature, wherein the Body does sympathize with the Mind, proves so beneficial, by the playing of the Muscles of the *Thorax*, and the pressing out of the saline Particles, so frequently and sensibly as it does in some chearful People, to whom it is more habitual than others, insomuch that it comes to be Proverbially, a Cause of Fatness. All these Instances shew, that Nature seems to receive more Relief from the Compression or acting of the containing Parts strictly taken, than most Men easily imagine.

As to extraordinary Efforts of Nature, to pass by what happens upon a sudden Surprise, Fear, Passion, or the like; it is certain, that Torture will raise the Spirits for some time very much, and there are as great Effects follow upon Irritation, which does not come up to the perfect Notion of Torture. What wonderful Effects do we see produc'd by strong Emeticks given by Surgeons in some Cases of the Limbs and extreme Parts? Where a Person so griev'd has oftentimes a robust Constitution and perfect Health, there the Medicines, tho' given inwardly, cannot be suppos'd to act after the common manner of Alteratives, by passing into the Blood, by the Spirituousness of their Parts, or the like; for the Person being in Health needs no Alteration to be made in the Blood, and other Juices, which are as good as they can be desir'd to be; but by the Irritation of the Fibres in the Ventricle, the Spirits are rais'd to the highest pitch they are capable of, and brought to communicate that Elasticity to the whole Body, all the Springs of Life are wound up, all the Pumps of Nature (if I may so speak) set a playing, and by these means the Agony is extended to the extreme Part affected, and the Matter fixt there is attenuated and brought to flow, that it may be absorb'd by the Blood, and discharg'd in the Circulation; Nay, we

may take notice of the great Power of a more gentle Irritation of those Fibres in those weaker *Hysterical* People, whose Spirits are of so fine a Make, or so scatter'd and weakened, that they can't long bear Fasting, without very troublesome Symptoms following upon it, for they, we may perceive, are in a sense strengthen'd at those times that the Contents of the Stomach happen to be so rarifi'd, as to cause a gentle *Ægritudo*, a lingering Sickness and Nauseousness, tho' not sufficient to cause 'em to vomit, for they shall then dispense with the want of that Food, without which at other times they could not possibly subsist with any tolerable Ease, and find themselves as strong and as free from their Tremors, Shiverings, and other ill Symptoms, as if they had eat and drank plentifully; and likewise during that Sickness the Salts shall come off plentifully in the Urine, which will then recover its proper Colour, tho' it was before as limpid as common Water; from hence it is manifest, that the Animal Spirits may be made to expand, dilate, or in some ways act upon themselves, without the encreasing their quantity by such internal Medicines as may be suppos'd to be converted into their Substance.

Having premis'd these things, I shall proceed to enquire, after what manner Nature endeavours to clear her self of some few Distempers, which I shall consider in their proper place, and likewise how she may the more easily succeed in those Endeavours, if duly assisted by moderate Exercise; which Assistance, if it at first View may seem too slow and gentle to produce so great Effects, will yet with the Allowance justly due to all sorts of Alterative Physick, viz. of a Habit or frequent Repetition, appear to be sufficient to procure those Ends I shall assign to it.

There is this Difference between the most compleat Productions of Humane Artifice, and that Divine Piece of Mechanism, the Body of Man, that the former are always the worse for wearing, and decay by Use and Motion; the latter, notwithstanding the Tenderness of its Contexture, improves by Exercise, and acquires by frequent Motion an Ability to last the longer; and tho' the Circulation, and continual and infinite Succession of Particles, are the immediate Cause of Life, yet the Health, the Strength, the Well-being of the Individual, is in great part owing to the Effects of a General Motion superinduc'd to these internal Motions; which it is so far from disordering, that it aids and assists 'em to a greater degree than we are wont to imagine; for in our Considerations of the Animal Oeconomy, we

seem to regard Nature only as in a quiescent State, without a due Allowance for the Alterations caus'd by the Motion of the whole, which yet are confess'd by all to be sometimes of great Consequence: for that General Motion acting both on the Fluids and Solids of the Body, may sometimes prove the last and best Resort for the Restoring the *Æquilibrium* between 'em.

As for the Fluids; One would think the Shape and Make of the Blood-Vessels were sufficient alone to lead us into an Opinion of the Necessity of Exercise; by reason they all terminate in a Cone, they must needs resist the Passage of the Blood incomparably more than they would have done if they had been Cylindrical; and tho' all the Branches of the Capillary Arteries, would, if taken Collectively, make a greater Diameter than that of the great Artery, yet the Consistence of the Blood, and the extreme Fineness of those invisible Meanders, require the frequent Pressure and Assistance of the Muscles to encrease the Circulation, which accordingly we always find very much augmented by those Means; yet 'tis the Result of this swifter Current of the Blood, which should be most valuable to us, I mean the better Digestion and Mixture of the various Particles convey'd into the Blood. I believe it will be allowed on all hands, that the best way to bring an Animal Fluid to a greater degree of Perfection, is Digestion; and the Excellency of that Operation consists in the just Degree of Heat which causes it; or, to speak perhaps more properly, in the just Agitation or intestine Motion of the Particles which may be suppos'd to occasion that Heat. The Standard or Measure of this Heat or Agitation in the Animal Oeconomy, is to be taken from what we observe in a Man in perfect Health, and in the Prime of his Age; when his Blood flows with its due Velocity, when there is an uninterrupted Secretion of all that is disagreeable to it, and it is wrought up to its florid Consistence, and a just proportion between the Serous and Grumous parts. Now this we may successfully imitate by repeated Exercise, when the Blood happens to be impoverish'd and Languid, we may encrease the Velocity of the Circulation, and consequently the Heat following upon it, by which a great many crude Particles will be attenuated and ripen'd, either for Mixture or Secretion, and there will be an equal Distribution of the attenuated Particles to the several Emunctories of the whole Body, by reason of the Solids co-operating with the Fluids; whereas it is often quite otherwise, when an internal Medicine is given design'd for one Secretion only, which may promote that, and perhaps hinder another; as a Medicine

which agrees with the Stomach, sometimes offends the Head; for the Nervous Parts being, as it were, Passive in the Case, the Secretions cannot be so equally performed as when the whole Body is exercis'd. I would not be here mis-understood, as if I suppos'd that this first Effect of Motion, this Digestion, would avail in many Cases, as where an ill Ferment is lodg'd in the Glands, or where the Morbifick Particles have been long a forming, and are strongly combin'd in the Blood, but this may take place where a greater degree of Agitation is absolutely necessary; as when the Blood is Effete and Languid, when the Chyle comes into it dispirited, and when even a proper Medicine proves offensive and burthensome, and there is scarce Power enough left in the Blood to master its Particles, and apply 'em to their proper Uses; then, I say, 'tis time to make the Solids assist the Fluids in the dispensing of this Load which lies so hard upon 'em, which by gentle and close Exercise may be more easily done than many imagine; besides there may be a Distemper occasion'd by Particles of a looser Texture in the Blood, than is usual in most Cases, where Nature may contend and struggle with the hostile Particles, and yet not be able to get the Victory; where there may not be a perfect Fever, nor yet a quiet Coalition between the Blood and those foreign Particles. In such a Distemper as this, it must needs be very proper to give a due Agitation to the Blood, to prepare those Particles for the several Emunctories that are ready to receive 'em: and this may be perform'd by a just Digestion, if we do but consider how much the Body is adapted to it, and how much more Noble the Digestions are in the Animal Oeconomy, than those produc'd by humane Contrivance. In all artificial Digestions the Particles which are to be separated by the Agitation of the Liquor, must either evaporate, or subside; but in the Body there are a multitude of excretory Ducts ready to receive the Particles, of such a determinate Figure, as renders 'em excrementitious, and proper to be cast off; so that nothing is left but what is proper to the Animal Fluids, and which the Vehemence of the Motion mixes and unites at the same time that it breaks and moulds the others for their proper excretory Channels; so that the Agitation is in this Case (as Dr. Grew very well expresses it in his Treatise of Mixture) as "*carrying the Key to and fro, till it hit the Lock; or within the Lock, till it hit the Wards.*" How do we know the exact Degree of Agitation, that is requisite to unite the Particles of the Fat, which are continually flowing in a very great quantity into the Blood, with the Aqueous, by the Means of sulphureous or saline Particles? Do we not

frequently observe in scorbutick Persons, who have lead a sedentary Life, that their Urines are cover'd with an oily Film of several Colours? and is it not very natural to suppose from thence, that the Blood wants a due Motion to keep those oily Parts united with the others? But it is no wonder, if these things are not well consider'd, when there is scarce any who makes Allowance enough for the quantity of the fat Particles, which are continually passing into the Blood, which must needs be very great, seeing the whole Skin is lin'd with its Vessels, besides what is heap'd up about the *Omentum* and the Kidneys; so that Unctuous Medicines are copiously intruded upon the Habit of the Body when there is a great Wasting of the Flesh, without regarding that the Blood is not able to master the natural *Pinguedo*, but gives it down daily, in all probability, for want of a just degree of Agitation or Digestion, to keep it suspended in the Blood, and to apply it to its proper Uses, and prepare it for its proper Vessels.

Besides the Power of Exercise on the Secretions of Particles purely Excrementitious, and the better Mixture of those which are Homogeneous; it is to be consider'd, that there are in the Oeconomy Secretions made to return with Advantage into the Blood, out of which they are made; and the Consideration of the Nature of these does afford us fresh Reasons to set a Value upon the Use of Exercise, because the Body is so fram'd and adapted, as to require it, in order to the furthering and increasing these Operations; and if in the Business of Fermentation, which is only a gradual Separation of the Must from the Spirituous Particles of the Liquor, we find that the Motion of the Vessels in which the Liquor fomenting is contain'd, does so much improve that Operation, as we are convinc'd it does, by the Effects of the Carriage by Sea on Wines and other Liquors in Casks; of how much greater Importance must the Motion of the Body be, in order to the perfecting the Animal Fluids, in a System of Mechanism so contriv'd, as to expect and demand such an Assistance? Where the Solids are so fine-spun, as to determine the very Shape of the Particles of a Fluid; and where they are so dispos'd, that a Fluid never passes by 'em but it carries off some Melioration and Improvement, and therefore cannot well arrive too frequently at those Passages where it receives so happy an Alteration. Let us suppose the Blood to pass the most extreme Parts twelve times in an Hour, when the Body is not mov'd; if the Motion of the Body encreases this to fifteen or sixteen times in an Hour, it will necessarily follow, that the Quantity of the Secretions by the Liver, the Spleen, the Brain, and the rest

of the Glands, which separate the beneficial Juices, of which I am speaking; the quantity of these, I say, must needs be augmented; which in Process of Time, when this is brought to a Habit, must be of some Consequence. To insist but on one of these Secretions; I take it to be no Paradox, that the more a Man stirs himself, the more Animal Spirits are made in the Brain; tho' it will be strait retorted, that by the very same Motion and Exercise, there will be a Waste of the Spirits by Perspiration, more than proportionable to the Overplus that is made in the Brain; and tho' I grant this, it will not suffice to discompensate the Benefit which the Blood reaps from the Augmentation of the Quantity of the Animal Spirits infus'd into it (if I may so speak) from the Brain; because the true Animal Spirits have their Work to do in the Blood, before they come to pass off at the Skin; they are not of that Fugitive Make, which at first Thought most Men are apt to suppose 'em to be; they seem to be destin'd to temperate the Acrimony of the Blood, to embrue it with a Plastick Quality, and may serve to execute other Functions, besides that of Motion; so that it is not at all to be wonder'd, if a Person, much accusom'd to Exercises, notwithstanding the daily Expence of a greater Perspiration, should have his Blood of a better Condition, and more Rich than that of another Person living a sedentary Life, by reason of the greater Impression, the greater Tincture (if I may be allow'd so to speak) of this most exquisite and inimitable Fluid.

These Things are not to be stated exactly, and yet they are not to be accounted altogether precarious; for tho' we shall never perhaps be able to know exactly what the Animal Spirits are, yet we may make a shift to distinguish what they are not. According to the common Notions, a well-prepar'd volatile Salt, after it has pass'd the Lacteals, and comes into the Blood, might be taken to be a pure Animal Spirit; and yet, undoubtedly, the Fluid, prepar'd by the Glands of the Brain, has something in it transcendently preferable to any thing that can be the Effect of Art. Whether those Glands are so dispos'd, as to unite some Nitro-Aerial Particles with others proper to serve as a Vehicle to 'em, is not to be determin'd by me or any Body else; but it may not be altogether so absurd, to guess at some such thing, since we know nothing in Nature that can afford Particles of that Elasticity as Nitre does; and we may discern, that the Animal Spirits seem to consist of a *Fulgur*, an *Impetum faciens*, something that is Irraditating; and yet withal there seems to be something extremely Mild and Plastick, and as it were Tenacious, combin'd with the Elastick. I hope I don't run into

an *Hypothesis*; I would carry this no farther than it can be kept in Countenance by *Phænomena* arising in the Cure of Distempers; for thus we see in the Nervous Atrophy, tho' the Spirits, taken as *Impetum facientes*, pass freely, and are not obstructed, as in the Palsie, yet the Benign Plastick Quality seems to be wanting, because the Habit of Body does not thrive, tho' the Spirits are brought all over it; and that the Spirits, when they are in their true Purity, are concern'd in Nutrition, is plain enough; because the intercepting of 'em, by cutting off a Nerve, always causes the wasting of the Part to which that Nerve lead.

Thus I have endeavour'd to shew some of the secret Advantages accruing to us from the frequent Use of Exercise; and by which it will appear, that the Fluids of the Body are of such a Texture, as will admit of Improvement from the greatest Rapidity of their Current, that is consistent with the Organs thro' which they are convey'd.

But, *Lastly*, to put these things past all Doubt, by a well-known Observation; we need but consider what is sometimes the Effect of too much Exercise upon taking a Purging Medicine, and that is an *Hypercatharsis*. The Particles of the Drug being sublim'd, and render'd more active by the greater Agitation in the Body, display a much greater Violence than otherwise they would have done in a Person of the same Constitution, who had given himself to Repose. This has been long ago observ'd by *Hippocrates*, in the *fourteenth* and *fifteenth Aphorisms* of his *fourth Section*; From whence it naturally follows, that the Motion of the Body may cause great Alterations in the Blood; may very much improve any Juices that are convey'd into it; and that in some nice Cases, where the Alterative Physick is very mild, and perhaps given in too small a Quantity, it is absolutely necessary to have Recourse to the Use of Exercise, to give an Energy to it, that it may produce the desired Effect.

Having thus briefly consider'd the Power of the Use of Exercise, I come now to shew, after what manner it affects the Solids; and that I take to be, *first*, by giving a greater Tension to 'em, or restoring the true Tone of the Parts, by curing the Relaxation by which they were weakned.

That I may explain what I mean by this *Tension* or *Tone* of the Parts, it will be necessary to consider, *first*, how great is the Benefit we daily receive by only changing the Position of our Bodies, from an Horizontal to an Erect

Position when we quit our Beds, where in the time of Sleep the Body has been relaxed; and this will appear best by the ill Consequences which follow upon a Person's being confin'd by an Accident to keep his Bed for a few days; for such a one always finds upon his getting up again, that his Spirits are disorder'd; he finds himself Vertiginous in some measure, and a great deal weaker than he was, before he betook himself to that Posture: From whence it evidently appears, that Standing or Sitting, the familiar Exercises (if I may so call 'em) of the most sedentary Life, are absolutely requisite to keep up the Balance on the part of the Solids, even in a State of Health, and that more violent Exercises are as requisite to recover this Balance, when sunk by Sickness. I know it may be objected here, that this Observation is chiefly owing to the Custom of changing the Posture of the Body alternately, in such a space of time, and that the breaking of that Custom occasions those Disorders; but this will not suffice; for an erect Position is essential to the well being of the Body of Man: and if the Infant was not at such an Age brought to it by degrees by the Nurse, tho' it might grow up to the Bulk of a Man, and live many Years, yet it would be a kind of bedridden Creature; Paralytick, as to the Use of its Limbs, tho' with the Sense of Feeling; and much weaker internally, for want of that Advantage, which the Fluids receive from the Solids by this most familiar degree of Tension; which we experience in Standing, which Posture has ever been esteem'd as a Tonick Motion.

But the Stiffness or Strength of the solid Parts will appear more evident by that sensible encrease of the Strength, which Men experience when they set themselves upon any vigorous Exercise; which continues, till thro' the Greatness of the Perspiration they grow tyr'd, and relax again; or, to borrow an Illustration from Beasts, It will appear by what *Jockeys* observe, who when they design to take the Bearings of a Running Horse, that is, measure the Extent of his Stroaks, they usually let him gallop a Mile or more first, as supposing that he can't come upon his Legs (as they term it) till he has run a considerable time; that is, he can't strike out so far, tho' press'd ever so much to it, upon his first setting out, as he can after he has run some time: which explains what I assert, That a proper or due degree of Exercise, enables the Nerves to dilate themselves sufficiently to take a greater Quantity of Animal Spirits, or some other way, to us unknown, gives 'em a better Tone, or Elater, and consequently fits 'em for more vigorous Actions.

But to bring the Sense of this Tension nearer to a Case of Sickness; Let any Man reflect, how he found himself after an Acute Distemper, wherein the solid Parts were mightily relax'd by the Heat of the Fever; when a Man in that Case rises first from his sick Bed, and makes a shift to walk a very little in his Chamber, tho' he quickly grows faint, and wants some Cordial to refresh him; that is, Tho' his Vital Spirits sink, as the Antients lov'd to express it, yet he perceives a certain Stiffness, Tension, or Strength in the solid Parts, by that first attempt to walk, which never leaves him, but encreases daily, till he recover his perfect Health.

These are some familiar Instances of the Sense we have, after what manner we come to acquire more Strength upon the Use of Exercise, and which every Man almost may recollect, that he has experienc'd more or less in himself; tho' in other Cases the solid Parts are always strengthening by Exercise, without so plain a Sense of it, as in the Instances above mention'd; as we see what excessive Strength some Men gradually acquire by a constant Practice of vehement Motions, begun when they are young, which growing upon 'em by degrees, they are not so sensible of the Encrease of it. This is the Case of Tumblers, Rope-dancers, and the like, in whom the Nervous and Solid Parts must be incomparably more wound up, more tense than in other People; and thus we see the strongest Men are often thin and Raw-bon'd, as we call it; that is, tho' daily hard Labour, and great Perspiration carry off a great deal of the grosser Fluids of the Body; yet are the Muscles not Flaccid, but Tense and Firm, capable of greater Actions than the Muscles of those who seem to have a better Habit of Body; which plainly indicates, that Exercise does communicate some Strength to the Nervous Parts, which cannot be any other way procur'd; and that we may argue from the greater to the less; that if healthy Persons may acquire such monstrous Strength by Use, People that are Valetudinary may, by setting themselves upon a resolute and diligent Practice of moderate Exercise, obtain a proportionable Increase of Strength.

It may be expected, perhaps, that I should endeavour to explain, how the Fibres come to receive a greater Power to act, by being often put upon Action, and to shew wherein the Elater, the Spring of the Solids, does consist; but this has been attempted so largely by an Eminent Author, Dr. *Baglivi*, and so much to the Disgust of very many, that it's better to rest contented with plain Experience, than to frame an *Hypothesis* for the *Modus*

of so abstruse a Method of Nature, which, in all likelyhood, Mankind will ever be ignorant of; but as far as we may reasonably guess, by frequent Distension the Nerves receive a greater quantity of Animal Spirits, because the Limb which is most us'd, grows biggest; and there is reason to induce us to suspect, that the Fibre it self strengthens by Use, has a peculiar Faculty to exert it self more and more, as often as the *Imperium Voluntatis*, the *Fiat of the Will*, sets it upon Motion. But unless we knew the Bond of Union, and understood how the Rational Soul acts upon the Animal Powers, we must be content to be most ignorant, the nearer we approach in our Disquisitions to that Union; but the Experimental Knowledge of these Parts sufficiently reproves those who hope to be deliver'd from some Distempers seated in the Solids, without acting suitably to the Nature of the Solids, the Subject of their Distempers: As for Instance; Suppose a Person, by frequent and unnecessary Use of the Bagnio, and more unnecessary Bleedings; by the Use of hot Liquors, and a perfect Disuse of all Bodily Exercise; by Passions of the Mind, and other Irregularities, is brought into the very worst of Hysterick Symptomes, with a Flaccidity and Relaxation of the whole Nervous System; How ridiculous is it, for such a Person to expect to be perfectly restor'd to a firm Habit of Body by internal Alteratives, and Methods little different from those things which occasion'd the Distemper; which tho' they may give wonderful Relief in the Paroxysm, yet can never restore the Tone of the Solids, which must be treated in a manner proper to themselves, by Frictions, Exercise of the Body, the Cold Bath, and the like; which are very likely to be able to succeed to a perfect Cure? For why ought we not to suppose, that as all Fluids have more or less a Tendency to purifie and exalt themselves by Fermentation, the Solids should otherwise have a Propensity proper to their Make, to recover themselves by a due Tension? And what can be more reasonable and natural, than to conclude, that if a Supine and Luxurious Course of Life has enervated the Body, an Active and Vigorous one should restore it? If it be objected, That gentle Emeticks have gone a great way towards procuring a perfect Recovery from some Hysterick Cases; I have premis'd already, that they act upon the Fibres, and put 'em upon frequent Contractions, much after the same manner as a total Exercise of the Body; and therefore in this last Case, where the first Passages are not in Fault, their way of acting illustrates what I assert, that the Solids must be made to strengthen themselves, and recover their Spring by frequent Endeavours. But for Alteratives, strictly taken, I

think it may be justly a Question, Whether there is that Medicine in Nature, that can remove this Distemper, when it has been of long continuance, tho' the World has been taught above an Age ago by *Paracelsus* and his Followers, to expect what I fear is not within the Extent of Nature; and the Impudence of Empiricks is so great, as to promise every thing that is absurd and Romantick; which keeps People up with hopes, that they may be so happy as to meet with that mighty Secret, which even in the quantity of a few drops, shall as it were charm away the most troublesome and riveted Distemper, and so instead of being Cur'd, they are Kill'd by Expectation; when the Power of Recovering their Health was in their own hands, if they had resolutely set about the proper means; like the Country-Fellow in the Fable, who when his Cart stuck fast in the Mire, must needs be calling upon *Hercules* to come and help him, when with setting his own Shoulders to the Wheels, he might easily have got clear.

But moreover, a second Advantage arising from Exercise, is, that it gives the Solid and Nervous Parts a grateful Sensation, which in some Cases is not contemptible; a gentle Agitation of the Spirits being able to remove some Pain situated in those Parts, which perhaps nothing else would remove so surely and so soon. To explain the manner of this by a trivial Observation (if any thing in Nature can be so) let us consider, how we can separate the *Cuticle* from the true *Cutis* without Pain; it can't be done with an Instrument without extream Pain, it can't be done by Vesicatories without some Pain; but it may be done with Chafing without any Pain at all, or rather with some Pleasure, till you leave off Rubbing, and the Air comes to act upon the naked Fibres. Now this can't be attributed to the Heat of the Part only, for then hot Medicines apply'd to the Skin; would do it as easily but must be suppos'd to be owing to a certain Agitation of the Spirits in the Extremity of the Fibres, which affects 'em with so agreeable a Sensation, as to surmount even the Pain of a Separation of their Covering, the Scarf-Skin. Now it will be allow'd by all, that whatsoever Sensation there is in the Extremity of the Fibre, the same there is at the Origine of it in the Brain; so that a pleasant Sensation in the Extremity, must needs be the same in the common *Sensorium*; and therefore 'tis easie to account for the good Effects of Frictions of the Limbs in some sort of Fits, by giving a new and different Motion to the Spirits, and thereby disengaging 'em from their disorderly Motions. And then if acting thus upon the Extremity of the Fibres produces such an Effect in 'em, 'tis natural to imagine, that that Motion, or gentle

Concussion, which much after the same manner, in some proportion acts upon the whole Body of the Nerves, must affect 'em with a Sensation proportionably agreeable, and may prove sufficient to dispose the Spirits to leave their Dispositions, and irregular Motions, when they happen to be so discompos'd, and consequently remove the troublesome Watchings and painful Symptomes occasion'd by those Dispositions, when other Means prove ineffectual. Thus we see how natural it is for those Hysterick Persons, who are vexed with obstinate Watchings, to fall into a true and refreshing Slumber, by the Motion of a Chariot, when Opiats will have no effect upon 'em, but rather encrease their Watchings: The Spirits being the most stubborn Part of the Animal Oeconomy, and not always to be compel'd, even by that potent Drug.

The abstruse Nature of this Part of the Animal Oeconomy, will not permit us to come at a fair Explication of these Phænomena, any otherwise, than by what we at different times experience; for we must first understand, as the Author of the *Dispensary* very well expresses it.

*How the same Nerves are fashion'd to sustain
The greatest Pleasure, and the greatest Pain.*

Dispens. pag. 3.

But we may discern, that the very Interruption of Pain is some degree of Pleasure; and that the lesser degrees both of Pain and Pleasure have something of a Relative Nature in 'em; a Person that is afflicted with some Pain, finds some Alleviation of it by tumbling and tossing in his Bed; which tumbling and tossing, at another time, would be a sort of Pain; and if any one thinks this ought to be imputed to Phancy, rather than Reality, I answer, 'Tis such a Phancy as none can be free from; and the denying a Person, in such Circumstances, the Liberty of gratifying it, would be the greatest Cruelty, and a high Aggravation of the Pain: Besides, in Pains of the Membranes, proceeding from the Corrosion of Sharp Humours thrown upon 'em, where the Part cannot but be very sensible of the Pain those Particles cause; yet even in this Case, the Spirits may be interrupted or diverted from the performing so acutely their Office of Sensation, by being

put into different Motions. I knew a hardy labouring Man, who hapning to be seiz'd with a violent Pain in his Hip, for two or three Nights, as soon as he came to Bed, kept beating his bare Hip with a Bed-staff a great while together, before he could get any Rest, and by that Means blunted the Pain, and tired himself into Sleep; (tho' afterwards he removed both the Pain and the Cause, by running a Packneedle himself thro' part of his Hip); now if thus much may be done in the Membranous Parts, where the Cause of the Pain is *ab extra*, what may not be done in Nervous Cases, where the Disorders of the Spirits are the prime Occasion of the Pains, if we can communicate to those Spirits, a Motion contrary to that Motion which occasions the Pains, which certainly may be done, by moderate and agreeable Exercise?

From these Considerations I cannot but be induc'd to think, that in all obstinate Pains, caus'd by the irregular Motions of the Spirits, and in the true Hysterick Colick, one of the most frequent of those sort of Pains; it would be more natural, and in no wise absurd, to recommend to the Patient, the Use of a Chaise, or light Calash, even in the Paroxysm it self, than the Fatigue of Medicines; the best of which, except Opiates, so often prove delusive. That Exercise is convenient for Women, with Liberty to sit or lie; and tho' the Motion at first may seem a little troublesome, and the Shocks too rude; yet I think, upon what I have hinted before, there is great Reason to expect, that after a little Patience the Spirits would be brought to relent, and disengage themselves from the *Plexus's*, where they occasion so great Pain. I am the more confirm'd in this Opinion, because there is a Pain which seems more deeply rooted, even in the Tendons of the Muscles, *viz.* the Cramp, which will frequently go off, by changing the Posture the Part was in, when it was first seiz'd; and especially by getting out of the Bed, and walking a little while, when no Pressure or Ligature will remove it, unless the Person rise; by which undoubtedly the Spirits are call'd back into some of the superiour Muscles, or some way or other put into a new sort of Motion.

This Opinion may be corroborated likewise by what has been experienc'd by some Hysterick People, who when they have lain perhaps half a Night restless and disturb'd, and without the least Inclination to Sleep, upon getting out of their Beds, and walking a turn or two about the Room, shall find themselves quite alter'd, and when they come into the Bed

again, sleep well; so that if so sudden and short an Alteration of the Posture of the Body, can produce so good an Effect, much more may be expected from the Exercise I have above mention'd; wherein the sick Person may at once enjoy the Convenience of a Cradle, and the Vehemence of a Exercise.

I might pursue this Notion, in considering the *Scorbutick Rheumatism*; in which Case the Persons afflicted are generally strong, and able to undergo any sort of Exercise; and therefore all the sorts of Exercise which I shall hereafter mention, will agree with 'em: But it will be needless to multiply words, that Distemper being chiefly seated in the Nerves, what I have said already will serve to illustrate the Advantage, which Persons griev'd with that Distemper might receive from a resolute and prudent Use of Exercise.

I hope these Observations on the Solids, will suffice to shew the Power of Exercise on this part of our Bodies; and if any of these Speculations may seem too nice, I would be understood, that I consider 'em as brought to a Habit, as frequently and closely repeated; not as the Use of Exercise is generally abus'd, being frequently undertaken, but seldom gone thro' with. 'Tis the want of a due Notion of a Habit, which has occasion'd the Neglect of this valuable Medium in Physick: Did People allow but the same regard to this, as they do to all other Alterative Physick, it would soon appear, how great Effects it could produce.

How ridiculous would a Man seem, who, when his Physician had recommended some Medicine to be taken to the quantity of a Drachm, or half a Drachm, should go and take half an Ounce of it, and then exclaim against the Medicine, that it disturb'd him, and did him a great deal of Mischief, and that he would never take it more: Or if instead of taking a moderate Quantity twice a day, for a considerable time, he should take that moderate Quantity but once in two or three days, and then exclaim that the Medicine was ineffectual? He that should act thus, would be thought to be a very unreasonable Person; and yet after this manner most sick People set upon the Use of Exercise. You shall have a Man ride fifteen or twenty Mile, when he should ride seven or eight, come home very much tyr'd, resolve never to be so serv'd again; and so perfectly lay aside all hopes of any good from the more moderate Use of that Exercise: Another shall ride out five or six Mile once in two or three days, finds no great matter of Relief, despairs of any Success from that Course, thinks it a trivial Thing, a meer Phancy, when the Physician does not know what to do, and so he wholly leaves off

too: Now allowing moderate Exercise to be a Medium for the Recovering our Health, this is a very unfair way of making use of it; for when once a Distemper will not be driven out by rough Means, by Purging and Vomits, but we must come to Alterative Physick, the Work must go on gradually, and that Physick must be us'd without Intermission. What is the difference between Aliment and a Medicament, but this? The first is chang'd into our Nature; the last changes our Nature. Now it would be as ridiculous for a Man to expect that gentle Drugs or gentle Means should alter his Constitution, if taken with great Intervals, as it would be for a Man to expect that the Bulk of his Body should keep up or encrease, tho' he eat but once in two or three Days; and whatever Regard is due to internal Alterative Physick, the same is due to the moderate Use of Exercise; for if by it the Secretions are equally promoted, and the Subject-Matter of the Disease brought to despumate slowly; it is highly requisite, that these Means should be closely repeated, with Moderation; that Nature may not be confounded and weakned, instead of being reliev'd; and without any irregular Intermission, lest the Springs should run down again; lest the Disease should have time to ruine faster than the Means of Cure can build up.

We see, by continual dropping, so soft a Body as Water can act upon a Stone; we see by incessantly following his Blow, the Smith can bring Heat into his Bar of Iron; so that where the Act it self, simply consider'd, is weak and trivial, yet the Habit is of the greatest Efficacy.

Neither ought this to discourage any, who will give themselves leave to consider, how slow, and yet how sure, some of the Despumations of general Secretions of Nature; are wherein, if the Certainty and Security will compensate for the Slowness of the Progress, they have Reason to acquiesce and submit, when there is no other Remedy left. How often has it been observ'd, that in some Paralytick Cases, after a considerable Use of the Hot Baths, the sick Person has gone away disconsolate, without any present sensible relief, and yet found himself cur'd in a Month or two after; the Morbifick Matter being just mov'd and brought to flow, when he left off Bathing, and yet not perceptible to himself; and if Nature can be enabled to make such real tho' flow, and for a time, insensible Advances towards Health, in a Subject half dead; may not we, with a great deal more Reason, expect the same and much more in a Person who has his Nerves free, the Use of his Limbs; and who, notwithstanding his Decay, is able to set upon a

Course of Exercise? If Men were not wanting to themselves in a Resolution to undergo with Patience the Fatigue of Reducing Nature indispos'd to its former State, by slow Measures, when violent are absolutely to be omitted; they would at last be really convinc'd, that Health, as well as Sickness, may approach insensibly; and that their tedious Struggles, and seemingly fruitless Endeavours did gain ground upon the secret and intimate Springs of the Oeconomy, before they come to be sensible of any the least Relief: for when once upon the use of such gradual Means, there appears a sensible Amendment, the Point is almost gain'd, and the Work more than half done: As we see that upon the Return of the Sun, after Winter, towards us, 'tis some Months before the Earth shews any great Signs of his Influence; yet when once it displays the Effects of it, we can very well discern, that they are such as must have been brooding long before we perceiv'd 'em. And why should not some Distempers go off leisurely, when we see so many come upon us so? There seems to be a Parity of Reason for it, tho' it is no very comfortable Consideration. We know the Poison of a mad Dog encreases in the Body for a Month or more, before it displays its fatal Symptomes; and there is a great deal of Reason to believe, that a Cancerous Humour is some Years ripening, before it creates any Trouble to the Person in whose Body it is bred; Why should it seem strange then, that some Diseases require a gentle and gradual Conflict of two or three Months, when perhaps they have been a longer Time growing upon the Patient?

What I have said would make the greater Impression, could we but have a History of the fatal Miscarriages which have hapned upon preposterous Methods of Cure; an History, which, I doubt, would prove a very Voluminous one; that Rashness being too usual in both Acute and Chronical cases; In the first, Many are apt to force an Indication, rather than wait for one. In the latter, The World abounds with Examples of the Folly and Impatience of Mankind. To instance but in the Dropsie; Who is there almost, who cannot furnish you with the Story of one, who, from a hopeful Condition in the use of Diureticks, and Corroborative Things, cast himself into the Grave, by violent Purgatives, recommended by some compassionate Friend or other, to carry off the Waters at once, with a Beadroll of Stories to vouch its Success; when the other Method, with a little Patience, had certainly brought him to his former Health, and perhaps in much less time than his Disease was contracted. So difficult it is for unhappy Man to bear the Penalty of some Months, for the Demerits of some

Years; and by Manly Consideration to keep from entangling himself in his Chain, instead of getting out of it.

I am not unaware here, how hard it is to frame Arguments that can have Force enough to prevail against the Apprehensions of the Pain and Trouble to be undergone in the first Attempt of Exercise, which most sick People have conceiv'd; and which are oftentimes so strong, as to blind the Mind, or bribe the Will and there is no way to deal with those People, but by Precedents; by shewing 'em, that those Difficulties have in many Cases been easily overcome: And here the Cold Bath offers it self, a severe Method of Cure taken up lately among us, and which upon the first Consideration carries Terrour enough in it; which if anyone had presum'd to recommend some Years ago, he would have been thought one of the most Wild and Barbarous of Men; and yet we see now the tenderest of the fair Sex dares commit her self to that terrible Element; and upon the first Experiment the Fears and Amusements vanish. How severe is the Sickness upon a Man's first going to Sea; equal seemingly to the Effects of any strong Poison; and yet Nature soon accustomes her self to that Motion which is the Cause of it, and the Sailor quickly grows well! And here it must not be suppos'd, that any salt Vapours arising from the Sun, do contribute to this Vomiting, for it is now well known to every one, who had but the least smattering in Distillations, That Salt will not rise with a much greater Heat than that of the Sun; besides it is observable that the oldest or most accustomed Sailors shall Vomit in bad Weather, when the Ship is put into an unusual and irregular Motion; so that it is plain, that the Motion of the Ship is the only cause of that Sea-sickness: if therefore Nature can so soon suit her self to a Motion that can cause such terrible Symptoms, how unreasonable, how Childish it is for any one to object against the Use of Exercise, because of the common and (in comparison of Sea-sickness) trivial inconveniencies which must be born in the first Tryals! Some strong People shall be confounded with a very few Glasses of Wine; and yet if those very People fall to keeping of Company, and addict themselves to Wine but a little while, they shall drink vast quantities without any Disorder. The first Pipe of Tabacco disturbs Nature to the utmost, but after two or three more, she becomes pleas'd with that, which before disturb'd her. In the Animal Oeconomy, every thing is so wonderfully contriv'd, and made to conspire for the Preservation of Life, that Nature can adapt her self to all Circumstances; she can expand her self to bear the Luxury of a

Palace, and contract her self to the short Allowance, the Bread and Water of a Prison; she can be easie under a Bloated Habit of Body, and she can make a shift to suit her self to the Expence of Fluxes and other Evacuations; accustoming her self so to bear 'em, that the longer they last, they may be in some Proportion the more familiar. But above all, she, with the most Ease, accustoms her self to the Use of Exercise; she may be said to delight her self in that, it being in a manner, *de Essentiâ Naturiæ*, and therefore it is in vain, when Exercise is really necessary, for a Person to complain after the first Tryal, and say, I'm tyr'd, my Bones are sore, my Head akes, I'm ready to faint, or the like; for all this must be endur'd, and upon patiently repeating the Motion, tho' no Abatement appear for some Days, yet the Reward will come At last: and as these Symptomes go off, the strength of the sick Person will encrease.

From these Considerations I think it sufficiently appears, that what I have before hinted, is not at all unlikely, viz. That in some Cases, a distemper'd Person may acquire, by suitable Exercise habitually us'd, a degree of Strength, as much greater, than that of other sick People in the same Circumstances, who wholly neglect all Exercise, as the Strength and Agility of Robust Men, bred up to Violent Motions, is greater than the Strength of other People, who tho' Healthy, yet are not us'd to such Things, and therefore incomparably Weaker.

Having thus Explain'd the Power of Motion, both on the Solids and Fluids, and having shew'd how necessary it is, that such Motion or Exercise should be continued to a Habit, that it may be render'd sufficient to procure those Ends it is directed to; I hope after these Considerations, it will appear pretty plain, that Exercise may deserve to be taken as a *common Aid* to Physick, (to use the Term which *Asclepiades* gave it) and 'tis under that Notion, that I propose it as so Beneficial a Medium in the Art of Curing; so that Exercise in this Sence is to Physick, as Bandage is to Surgery, an Assistance or Medium, without which, many other Administrations, tho' ever so Noble, will not succeed. It is a kind of Reserve, but yet of that Efficacy, that the thing you most depend upon, and tho' in it self very powerful, may yet receive its *Derniere Puissance* from this Reserve. And to this it is that we most undoubtedly attribute the wonderful Success which the Antients had in their Curing with such indifferent Materials, as their Pharmacy afforded 'em.

This will prove an Aid in a double Respect, viz. both of the Distemper, and of the Medicine.

In Respect of the Medicine; It is to be consider'd, that some Medicines may require it, to enhance their Virtue; others to remove some Inconvenience attending their Operation, which may deter People from using 'em so liberally as they ought to do.

As to the Former, the ordinary Circulation of the Blood, may not suffice to Answer the Nature of some Medicaments, and call out their utmost Efficacy; just as we see the heat of our Sun will cherish and keep alive some Exotick Plants, but yet will not suffice to bring 'em to their utmost Perfection, to flower and seed; so that Exercise in this Case, is like the just and exact Incubation to the Egg; that which Animates the Drug, and gives it a Power to produce the Effect it is directed to. A Medicine may not avail any more without Exercise, than Exercise without a Medicine, and yet when both are us'd together, there may be a Result from that Union, of the greatest Importance.

Therefore, before I come to speak of the Distempers, most liable to the Power of Exercise, I shall take Notice of two or three Remedies, which seem to demand this sort of Assistance.

The First, is the Decoctions of Woods; it is the general Complaint of those who take these for any Time, that they pall their Stomachs; to obviate which, if it be requisite that a Person should persist in this Course, nothing can be more proper than Riding, or some other gentle Exercise, because it will keep up the Vigour of the Spirits; and how much the Appetite depends upon that, is easie to imagine, besides that the Intention, the *Diaphoresis*, is likewise promoted thereby.

Another Medicine which should be followed with Exercise, is the Chalybeate, especially in Dropical Subjects; not for fear it should lye heavy upon the Stomach, as the Vulgar think, but because in these People, the Contents of the Stomach are much rarefi'd and flatulent, and the Steel is apt to cause Distentions and Gripes, and other troublesome Symptomes; so that it is necessary, the whole Body should be well warm'd, that those Particles may be discuss'd, and the Stomach qualifi'd to bear the Chalybeate; besides, that acquired Heat will enable it, after it comes into the Blood, to display its Effects the sooner, either as a Corroborative, or a Diuretick. In

Hysterick and Hypochondriacal Persons, this Medicine, gives trouble after another manner, by Costiveness, by Head-ach, and Heating the whole Body too much; now all these are much qualifi'd by Exercise, for it will procure a Ventilation of many of those Particles, which the Medicine agitates and throws upon the Membranes.

I might proceed to enquire into the Nature of *Balsamicks*, but that I shall have occasion, as I proceed, rather to say something against their Use, in one of the Distempers, which I shall consider; but if they are to be us'd, what I have already said in Relation to the Fluids, will shew that a great deal depends upon a proper degree of Agitation in the Blood, for the uniting and throughly mixing the Particles, of a Medicine of this Nature, that it may be transmitted to the designed Part to some Purpose; and as it would be convenient a Balsamick should be taken in a larger quantity, if the Stomach of sick People could bear it; so during the Time of Exercise, while the Body is heated, the Stomach can bear a greater quantity than at other times, without any Sense of Irritation, or Inclination to throw it up. But I shall forbear to enlarge any more on these things, and go on to the Distempers, which seem most Naturally to demand this kind of Assistance; in Treating of which it will be easie to discern in every several Case, how the Gymnastick Part will agree, or fall in with the Pharmaceutick.

OF THE *CONSUMPTION.*

The First of the Distempers then, is the *Consumption of the Lungs*; I take this to fall under the Power of Exercise; for these two Reasons.

First, Because the Morbifick Particles, which are the immediate Cause of the Disease, seem to be of a looser Texture, to be less intimately combin'd in the Blood, than in most Chronical Cases, the Particles which occasion each Distemperature seem to be.

Secondly, Because this Case requires the carrying off the Acrimonious Particles, by equal Secretions, rather than by any one particular Emunctory of the Body.

The *First* Reason seems to appear manifest enough, from the habitual Heat and Disturbance, which are generally complain'd of, sometimes even upon the first Breaking out of the Cough, and from the continual Quickness of the Pulse; all which shew, that there is an imperfect struggle of Nature, frequent and partial Ebullitions, which don't arise to a degree sufficient to clear Nature of that which oppresses her; but yet plainly indicate, that the hostile Particles do not unite, or accord with the Blood, so much as the Particles of each Disease do in other Cases; as for Instance, in Scrophulous and even in Cancerous Cases, tho' the Blood is loaded with so pernicious and even corrosive a Humour, yet we find no Disorder in the Beat of the *Artery*, no irregular Heats, but for some Reasons or other, in the make of their Particles, they pass better with the Blood, and the Disease is longer protracted; now I think it seems to be a Natural Consequence, that where there is an Ebullition or Contention of Particles, there is no Union; and that a more general and natural Heat, superinduc'd by Exercise, by the Solids acting uniformly upon the Fluids, may produce a Ventilation of many of those Particles, which Nature contends so much with.

The *Second* Reason, viz. The Necessity of equal Secretion, is occasion'd by the Effects of this Hectical Disposition, which by bringing a Languor upon the Spirits, a Relaxation or Flaccidity of the Muscular Parts, and even

of the Lungs it self, renders Nature unable to bear any particular Secretion without great Disturbance: Thus we see upon the use of the gentlest Purging Medicine, the Cough is encreas'd, and the whole Body for a Time, more than ordinarily disturb'd; the same happens upon the Use of *Sudorificks*, and indeed scarce any particular Secretion can be considerably enforc'd, without some Inconvenience following upon it; so that it must needs be the most proper Method, if we can attain to it, to enable Nature to do the Work her self, by gentle and even Despumption, of the acrimonious Particles, at all the Emunctories.

To procure this good Effect I propose the first of those Exercises, which I shall consider more amply in its proper Place, which is Moderate Riding. This Exercise is undoubtedly the most likely to cause an equal Exaltation of the Fluids, to restore the Tone, and Elasticity of the Ducts, so that the hot fretting Particles may be cast off; some of 'em by insensible Perspiration at the Skin, others by the Kidneys, others by the many *Salival* Glands, others by the Glands of the *Intestines*, where the very acrimonious Particles, forc'd out by that Exercise, which in a special manner acts upon those Parts, may be very much alter'd while they lye in the *Intestine*, undergo a sort of *Cohobation*, and in all likelihood may some of 'em become inflammable, and so dispos'd, as to prove Nutritious, when suck'd up into the Blood, as some of the Contents of the Intestines always are. This is communicating, *ab extra*, a Power to Nature to act upon her self; which must needs be more agreeable than to put a Force upon her, when she is Languid, and not able to master both the Drug and the Distemper.

It would be of great Consequence, to People Afflicted with this Distemper, if they would be brought to consider seriously the Distinction of the Oeconomy into the Parts containing, and the Parts contain'd, that is the Solids and Fluids, and the happiness of being able to Exert the Strength of the Solids, and make the Muscular and Nervous Parts assist the Blood and Spirits. There are Distempers wherein a Man is so Unhappy, as to have one Part of himself only Passive; as in Fevers, the Intenseness of the Heat, affects the Spirits and Nerves to that Degree, that all Power of Standing or Going is taken away. In a Palsey, the Hopes lye all in the Fluids or Liquor Contain'd: In other Cases, the larger Glands are so much alter'd in themselves, that the Motion of the Body would be to no Purpose; but here in this Distemper, we are Treating of, the Case is quite otherwise, if the Sick

Person will but Entertain a Resolution to help himself, will employ all the Springs and Fibres of his Body, and by that means take the Labouring Oar from lying always on the Blood alone, he will have no Reason to despair.

Thus I have consider'd how the Use of Moderate Riding will conduce to the conveying off the Subject matter of the Disease. The next Indication is the Strengthening the Tone of the Lungs and Muscular Parts, which in this Distemper grow Flaccid, I might add of the Stomach too, but that we can help that Bowel by many excellent Internal Remedies. Now I would fain know of any Man, how we can reach the Flaccidity of the Lungs, by Internal means, till the Distemperature of the Blood is remov'd, when it will go off in Course, but would be done much sooner, if we assisted both the Solids and Fluids at the same time; now that the very Lungs itself may appear, not to be out of the reach of a Habit of Exercise, let any one consider the strength of that Part, which Divers acquire by frequent Diving; or to come nearer to our Purpose, take any two Men equally, us'd to Hard Labour, of equal strength as near as we can guess whereof one has accusom'd himself to Running, the other never done so, all the World knows that the Practis'd Footman shall Run a great deal farther, and much faster than the other can do: Tho' in the Common Sense of the Expression, this latter has a Clear Wind as we say, and is in perfect Health; which invincibly proves, that the Lungs tho' a Bowel, are capable of a Habit, and that with a Proportional Allowance, the gentle, easie Exercise, of Riding, must introduce a New Habit, into the Lungs of a Consumptive Person, and so recover the Tone of that Bowel.

I know it will be reply'd here, that *Balsamick* and healing Medicines are suppos'd to strengthen the Parts they are directed to, that they are generous Medicines, of fine Parts, and consequently fitted to Communicate a firmness, a Spring to the Nervous and Membranous Parts of the Lungs; and if it could be prov'd that they did Heal so much as they have been pretended to do; I would readily allow they did Strengthen those Parts, but I have had some considerable Opportunity, to observe the Use of those Medicines, and I never could find that if Alteratives fail'd, Balsamicks would do any great good; that is, taken strictly as Balsamicks, upon a Healing Intention. I doubt not, but in the beginning of the Distemper, as Alteratives they may be of Service, especially the milder sort; by the pleasant sensation they Create, and the Consent of the Parts they will give present Abatement of the Cough,

and when brought into the Blood, may by Promoting a *Diuresis*, or by precipitating some of the Acrimony, help to carry off the Cause of the Cough, after the Alterative way, but that when there is any Ulceration in the Lungs, and the Blood is loaded with Hot and fretting Particles, they should then heal so much, I cannot conceive. If we will but give our selves leave to examine a little closely how they act, when externally apply'd to a Sore, we shall not perhaps find, that they are all of 'em such immediate Healers; some of 'em are too fine and Stimulating to be us'd as Eupoloticks, but rather prove Digestives, and therefore must be more likely to cause a too great Agitation in the Blood of these People, than a healing of the Ulcer; I know it may be here reply'd; that they are very proper to cleanse the Ulcerated parts of the Lungs in order to their better healing; but I can't imagine how it should come about, that there should be such great need of cleansing the *Ulcuscula* in a part of so Spongy and Membranous a Substance as the Lungs, where there can be no redundancy of Parenchymatous Juices to feed the Ulcers; besides it is to be consider'd, that the constant Motion of the Lungs, will help so deterge the Ulcerated part, just as if we should suppose a Man, that has an Ulcer in his Leg, should be squeezing the Lips of it together all day long, we can't doubt but he would by that means work out the *Pus*, the Slough, and all the mispurities of the Sore, and in like manner, the Heaving and Subsiding of the Lungs will hinder any thing from Bedding or Lodging it self long in a part that is really Ulcerated. And alas! here is the grand difficulty in a way to a Cure, we can't easily bring so arid a Substance, as that of the Lungs to unite, when lacerated, because of its continual Motion; so that there is all the reason in the World, for us to heap in only healing Medicines, strictly taken, without any thing that may prove in the least stimulating. Therefore, wherever Balsamicks have done any great good; I cannot think it has been any other way, than by deriving of the Acrimony from the Blood, and not by immediately healing the Part affected; so that tho' these are Noble Medicines in Colicks and Simple Affects of the Stomach, where the State of the Blood is quite different, yet here they are too Generous. They are like the Sword of a Gyant, in the Hands of a Dwarf, that will not help but Oppress. And as for the Oily Medicines, which may be call'd a sort of milder and Artificial *Balsamicks*, we ought to consider, that the Blood is Replenish'd with a better Oyl than any we can immediately supply it with; I mean the Fat, which to the quantity of a Pint at least is continually passing,

into, and out of the Blood: And yet in this Ill Habit of Body it wastes daily, and does not Unite with the other Fluids as in a state of Health. What then can we do by the Poor Addition of a few Drachms of Unctuous Stuff, which after it has pass'd the Stomach enters the Blood, to the quantify of a few Grains, and does not the least good, in Reparation for the unpleasantness in the Taking, and the Uneasiness it sometimes causes in the Stomach of the Sick Person?

I hope these Reflections will not be misinterpreted, as if I endeavour'd after some little Hypothetical Notion as a wedge to make way for any Design of mine; they will appear but too real to any that have been Conversant with this Distemper. I could wish it was all Hypothesis and Fiction, and that these Medicines would perform all that is expected from 'em, but then, to what must we attribute the Ravage this Disease makes, which is known to all, to be a Melancholy truth? Not to the want of *Balsamicks* certainly, for both Poor and Rich, can make a shift to procure enough of 'em. The Lozenge and Linctus are in every Bodies hand, but this must be attributed to their leading People, to take a wrong Aim, to level at the Symptom instead of the Disease, these specious Medicines induce 'em to be intent on the Cure of that, which is most Troublesome *viz.* the Cough, when they should lay the Ax to the Root of the Tree, be more intent on the Cure of the Habit of Body, and not let it be overrun with a Poisonous Acrimony. I am confident Legions of the Dead might have been above Ground, if they had but conceiv'd the Fallacy of these means, if they had but stuck close to the proper Quantities of any one good Alterative, they might have Plung'd out of their several Maladies; but by placing all their Hopes in things directed to the Cough, they have far'd like the Dog, which bites at the Stone that is thrown at him, instead of Biting him which threw it, not knowing that such diligent plying of these Medicines is a kind of Embalming a Man before his Death, and an Ill boding Presage that in a little time, he will be in a Condition to be Embalmed after it.

From what I have said it is plain, that I take the Negative way (if I may so Speak) of Curing this Disease, to be the most rely'd on, that is, the deriving the Acrimony, which causes the Cough and other Symptomes to the several Excretory Channels, and clearing the Blood of it; for the Blood when freed from such Acrid Particles will prove the best of Balsams it self. Therefore the milder Antiscorbuticks, the Bitters, Decoctions of Woods, and

even the milder Balsams, do all contribute their Assistance upon this Intention, in the first State of this Disease, and do very often secure the Person that makes use of 'em, and when they have not prevail'd alone, if the Use of Exercise had been superadded to 'em, they would undoubtedly at that time have been render'd effectual. But yet I am not so bound up in an Opinion, but that I am convinc'd there is such a thing, as a positive relief in this Case, in the strict Sense of the Expression; that is, a Healing of the part fretted or Ulcerated, but then I believe, it must be done by things of a milder Nature, than our Common *Balsamicks*. The Waters of our Hot Bath, are able to do a great deal, by the Healing Ores in which they abound, and there are other things which seem qualifi'd for this end; But that Qualification necessarily supposing they should be extremely Mild and Temperate, and upon the account of that Temper, it being likewise possible they may sometimes miss taking Effect; it is these considerations, have induc'd me to apply the Assistance of Exercise to the Temperament of those Medicines, that by such means they may be render'd able, always to Answer expectation. But that both the Nature of the Medicine, and the Assistance of the Exercise may appear the clearer, it will not be amiss, to consider two or three of these Medicines.

The first of 'em is a vegetable which has always been accounted a Pectoral; but after the Rate we use it, I much question whether it may not be said to be wholly indifferent; this is *Coltsfoot*, a Plant seemingly dry, and little likely to effect what I have known it do.

I shall here venture to give a Relation of some of the strange Effects of it, which are so seemingly incredible, that if I had not full Assurance of the Fact I should not offer it, and tho' it is not of a Cure of the same Distemper, which I am treating of, yet I hope it will not be thought a Digression, because the Obstinacy of the Humour, which is the cause of that Disease, which this Herb did remove, is so much greater than in the Case I am upon, that it may serve to give us Reason to expect great Relief from it, in the Cure of the Consumption likewise, to which it has always been apply'd, if us'd after the same manner, and in the same quantity, as it was in that Case, it was therefore a Scrophulous Subject that it reliev'd, but one so Deplorable, that the Hospitals can't often shew the like. The Young Gentlewoman had above twelve Sores upon her, she had had the Regular help of Physicians, but was left off as incurable, when a Person who was no

Physician, and did not pretend to any thing like dealing in Medicines, only he had reason to know the neglected Virtues of this Plant, came accidentally to the House, when the Gentlewoman's Mother was Lamenting her Daughter's Condition; after having given her Reason, to expect something from his Medicine, he promis'd to make it for her, but made her send 10 Miles, twice a Week to his house for the Decoction of the Herb, that he might conceal it from 'em, because he knew they would undoubtedly despise it, if they knew what it was: He therefore made very strong Decoctions of it, till the Liquor was Glutinous and Sweetish, of which she was to Drink as much as she could every day at what times she pleased, this she followed above four Months; in which time most of her Sores were dry'd up, and in a little time more, she was perfectly Cur'd. And of this I have reason to be certain, because I liv'd in the House where it was made, all the time, and the Person who made it, did not make a Secret of it for Gain, but only that it might not be slighted. This instance I have thus amply related, that it may serve as a hint that this Herb when it is us'd as a Pectoral, ought to be us'd after another manner than we generally do. And that when we do make use of Vegetables, in a manner suitable to their Nature; we may find Cause to come to a Temper, as to our Opinions concerning 'em, notwithstanding the great Plenty of generous Medicines, which Chymistry affords us. I have caus'd the Decoction of this Herb to be made after the same manner, and have given it where I did not expect a Cure, and thought that I had reason to believe, it did in some Measure prove Nutritive. And we find by *Reusner* in his Observations publish'd by *Velschius*, that it has been us'd as an Analeptick, he tells us that *Hillerus*, the Marquiss of *Brandenburgh's* Physician, did restore Children out of *Atrophy's*, by making 'em eat of this Herb fry'd after the manner of Clary.

The next thing I shall take Notice of, as peculiarly adapted to this Case is *Liquorice*. This Plant was ever reputed by the Ancients for the greatest quencher of Thirst in Nature, and therefore they call'd it *Adipson*, and upon that account, *Galen* tells us it was given to Dropsical people, *Theophrastus* calls it *Scythica*, and *Pliny* gives us the Reason of it, and tells us the *Scythians* where wont to Live 12 Days upon *Liquorice*, and a little Cheese made of Mare's Milk; so that it was in Reputation, likewise for sustaining Nature, and enabling People to bear Hunger. Its effects on Pains in the Stomach, the Bladder and the like, are numerous; and some of 'em very well attested, and perhaps there is scarce any Alterative that the Ancients

take more Notice of than this, except their admir'd *Silphium*; and we may gather from all, that it is one of the greatest Correcters of Acrimony in general, and that it is very temperate and safe, because the Juice of it has been drank in considerable quantities, and that fermented too; after this account of it, let us see how we use it; we boil about an Ounce or an Ounce and a half, in a Decoction of a Quart or two with other Ingredients; this is a wonderful Concession, but then in our Lozenges, there we do it to some purpose, about equal Parts of Juice of *Liquorice* and *Sugar*, make up a Stupendious Medicine indeed, not remembring at the same time a good Remark of *Tragus's*, viz. that *Sugar* and *Liquorice* are directly contrary, he Glories, speaking of *Liquorice*, that we have found a Sweet, that will quench the Thirst, whereas most other Sweets will cause Thirst, and instances in *Sugar*, which if it be true, can any thing imply more of Contradiction than Our Practice? If we were to make Sweetmeats for Children only, it would be allowable to mix all the Sweets in the Universe together; but when the Blood of a Poor Consumptive Wretch, is heated and loaded with Acrimony, to spoil the most agreeable Drug in Nature, by mixing it with its contrary, only because the form of a Medicine requires it; this, with all Submission, is what I think cannot easily be excus'd; this is to Cheat People with the *Bellaria* of Physick, and Tickle Men into the Grave.

I know what will here be the Objection viz. that these things are design'd only for the Cough, and not expected to Cure the Habit of the Body, and that therefore they may be allow'd to be a good sort of Composition for that Palliative Service they are directed to; but this will not suffice, for there is not one in ten that makes use of these Medicines, but relies on 'em for the Cure of the whole Distemper; and therefore this is the broken Reed that has deceiv'd so many; especially of the Poorer Sort, and which leads 'em in such numbers into the Hospitals to end their Days there, after they have lost the Opportunities of Recovery by depending on these Trifles. And if any one must needs take offence at some of these Expressions, let him consult *Ludovicus*, an allow'd Judge of these matters, in his *Pharmacia Moderno seculo applicanda*, he will find what is his Opinion of these things in the 19th Page of his *first Dissertation*, speaking of the *Confectiones communes & Candisatæ Conservæ recentiorum siccae (simplicis sui Pulvere plerumque debiliores) Martis Panes, Pandaleon, & antiquariæ Saponeæ Confecturæq; reliquæ*, he says, *Væ Hecticis tabidisq; quando tandem ad ejuscemodi Refectiva, sesamo atque papavere sparsa, pineis Pistaceis, &c. damnantur:*

Arentes hinc fauces (quamvis difficulter interdum) lenitas vidimus, curatum neminem, quin potius intensiores inde depascentes febres, dejectum magis appetitum, festinatosque Fluxus colliquativos. And speaking before, pag. 9. of *Decoctions* and *Infusions*, he says, *Procertis interdum Circumstantiis in Pectoralibus & Vulneraris dilutiora hæc contractioribus dosibus commodiora deprehenduntur*; and, it seems, he thinks this Observation, relating to the Use of *Pectorals*, to be of such Moment, that he makes it one of the Heads of his *Additionary Comment* or *Appendix*, where, pag. 582. he has these Words; *Natura interea nihilominus, præ Essentiis Extractisq; pectoralibus, præq; fauces in internis ibi ardoribus tantisper lenientibus Morsulis, Trochiscis atque mixturis antihecticis, antiphthisics, diffusius quidpiam & ad remotiora perveniens unà ut plurimum velle videtur.*

These Citations plainly shew, that he thought those sugar'd Compositions no apposite Remedy for Persons in such Circumstances, but that whatever Remedy is made use of, it ought to be made to dilute as much as possible; which does agree with the Reason, which I shall shew anon, for the plentiful use of those mild Vegetables. I have made these Citations at large, that what I have said may not be thought to be any Figment of mine, but that I may appear, that I have Precedent as well as Reason on my Side. But to return to the Root I was upon—Besides the mixing of Sugar with Liquorice, to what purpose is the Aqueous part of its Juice exhal'd; what harm would that soft *Lympha* do to People, who have a continual Thirst upon 'em? To what purpose must the Juice be inspissated, in order to acquire an Acrimony by lying, not to speak of its Adulterations? These are things which I could not forbear animadverting upon, because they put us out of the right use of a Medicine, than which there is not perhaps a greater *Analeptick* to be found, if it were taken in the same quantity as other Juices are taken. A Medicine that is a kind of a Balsam *in Ficri*, and the most likely to be wrought up to Perfection in the Blood, and of which the Fresh Juice ought undoubtedly be taken to a Spoonful or two several times a day. But thus it is, we give a thing the Name of Physick, and then stand aghast at it, and take it with Guard and Circumspection, as if it were not possible that any thing should prove a Medicine, and yet be taken in an Alimentary way.

There is another Plant, the *Cynogloss*, which seems not unlikely to be of Use in this Case, because it seems to have something of a like Gleamy Substance in it; it has been deliver'd down to us under some mistaken

Notions, as if it caused Sleep, which perhaps have been occasion'd by its Cooling and Styptick Quality; but a late Author of unquestion'd Judgment and Experience has us'd it pretty much in Decoctions with *Turnips*, and says, it has no such quality, but recommends it to People in this Distemper; to these may be added some of our Vulneraries, of which there is great Variety of all Rates, of all degrees of heat; and among 'em one of the Temperate sort, never enough to be valu'd, viz. the *Comfrees*, and which in Consumptions, upon spitting of Blood, may be expected to do great things; These Roots may be so manag'd by a good hand as to be eat as Food. The Female Retailers of Physick would perhaps take it Ill, if among these things I should forget their Preparations of *Turnips* and *Snails*, which may all have their time of being serviceable, either as Food or for Variety, and what is more, all these things are Compatible with a Milk Diet too; these things may be taken in small quantities at different times from the taking of the Milk; tho' if taken with it, they could cause no Coagulation, and so a mild and Medicated Chyle may be continually passing into the Blood to the great Advantage of the Sick.

These Instances are sufficient to shew the Nature of those things, which I take to be the most adequate Remedy in this Case; viz. that they ought to be such as are of a Medium, between common *Balsamicks* and *Acids*, and that they are such, as seem most likely to prove Nutritive to People in so weak a Condition; the reason why I set such a value upon these moderate things, is taken from the state of the Blood of People in such Circumstances, which seems unable to manage stronger Medicines, the least tendency to a *Diaphoresis* being some disturbance to those Persons; so that what is to be done, must be by things which may suit with the Blood, and as it were grow upon it, that may be transubstantiated into its *Crisis* after an Alimentary way; there must be a continual Rill of these temperate Juices into the Blood, without the observing of Physical Hours, and then 'tis to be hop'd the Blood may renew by degrees, and the Acrimony may decrease for want of Fuel; and thus we may perhaps better obviate the Periodical Ebullitions of the Hectick, by subtracting their Cause, than by stifling the Hectick by keeping in the Cause; I have not Scope here to explain my self, but I think the common Causes assign'd for those Fits, don't seem sufficient; I can't think the Ripening of a Tubercle able to do so much, that little quantity of *Pus* can't contain a *Putredo* sufficient for such effects, not to say the same Hectick happens, where no Tubercle has broke; to be short, it seems to me

most probable, that when the Blood is so much saturated with disagreeable Particles, as in Consumptive Persons it is, as these Particles encrease and grow upon those Particles which make up, the Proper, Genuine, Inseparable Essence of the Blood in its true, State; I say as the first gain ground, there is so great a Correspondence and Harmony in the Oeconomy, that these latter must contend and resist the other, tho' in the Contention, Nature gains no great Advantage, but only fights and retires till she is quite overcome; this seems to me no unlikely Idea of the Hectick, and if it be true, the best way must be to substract the quantity of the Morbisick Particles, by using such a Food, as cannot possibly afford Matter for 'em.

Having, then consider'd these Medicines, I will suppose it granted me, that they are proper in this Case; I won't say that they shall be Sufficient to Cure of themselves, (tho' I don't doubt but they may in some Constitutions do the Work themselves) but I will only suppose, that they do greatly dispose towards it, which *Postulatum* will, I conceive, be readily granted me; I will suppose likewise, that Riding (the Exercise I propose in this Case) does likewise dispose towards a Cure, which *Postulatum* will be granted too; I will suppose farther, that these two Courses are Compatible, and may be us'd together; as the Medicines help the Fluids, the Exercise helps both the Fluids and Solids; which *Postulatum* cannot be deny'd me neither; what then naturally Results from this, but that they be both us'd in Conjunction? And is it not more than probable, that these two Methods joyn'd, shall effect that which neither of 'em can singly? Do not we see enough of this every Day in Natural Occurrences, where one, two or three things, indifferent in themselves, shall, when blended together, produce a valuable Effect, which none of 'em could alone? And shall these things be observ'd in lesser Arts, and be slighted when a Man's Health is at Stake? Seeing we abound so in Compound Medicines, why may we not for once take up with a Compound Method of Cure, (if I may so speak) that is, if we cannot obtain Health by one sort of means alone, why may we not expect it from a Complication?

Thus I have run up these Arguments to a Head; I have shewn that the Medicines appropriated to this Case, ought to be very Mild and Temperate, upon the account of that less prevalent quality, there may be Hazard, lest they should not always prove equally effectual; and therefore to supply any such Defect, I substitute a most easie Natural Gymnastick Course, as a

common Aid to the weakness of the Medicines, and an assistance to that part of the Oeconomy, which those Medicines can't reach. Whether this is not most suitable to, and consistent with the even Tenour of Nature, tho' it may not relish so much of the Magnificence of Art, I must submit to those who are best Judges; to me it seems to promise enough, and carry more Healing with it, than some things that are dignifi'd with the great Titles of *Gilead* and *Peru*.

If after all there are any People who will think, I have taken too much upon me, in venturing to attack the *Balsamick* Method, if they cannot think slightly of Medicines, which will give such present mitigation of a Cough, and which are so Fragrant and Costly, let 'em enjoy their Opinion, and persist in the use of them; and if they find 'em at any time not so effectual as they could desire, let 'em but superadd the Power of Exercise, and they will doubtless find 'em much improv'd; and if they come by that Means to succeed, I shall not envy their good Effects.

Besides these two main Indications, there is something more to be consider'd in the Cure of the Consumption; and that is, how we may obviate the Moisture of the Air; which is a very troublesome Enemy to Consumptive People, of what Constitution soever, who dare not make use of Generous Liquors to fence against it; for that Practice would be prejudicial upon another Account: Now what can be more Natural in this Case, than the raising the Spirits to resist this Moisture, by a gentle Motion of the whole Body, which at the same time, causes a greater Degree of Heat, and that equally diffus'd all over the Body, which must needs rarify in some measure, the moist Air, and besides, make the hot and acrimonious Particles in the Blood, supply the place of warm Internal Medicines, which in another Person would have been proper to have been given, to oppose the Moisture of the Air? Now this is much the same, that the Change of Air can effect in the Body of a Sick Person, for 'tis the equal Influence, the universal moderate Rarefaction of a warm Air, that makes it so beneficial, and if we will cast in the benefit of the Tension, which is caus'd by moderate Riding, together with the Equality of the Heat, it will appear to be very little short of what is usually expected from a Journey into a Foreign Air, and I could here give an Instance of a Gentleman, who, when he was in the South of *France*, found but little Relief, any longer than when he was on Horse-back; and who after his return to *England*, found that Riding

supported him as much, as the Change of Air; So that upon the Consideration of the equal promoting of the insensible Perspiration, and the Benefit, which at the same time accrues to the Solid Parts, this Exercise which I have so much insisted on, may be allow'd to be almost, if not altogether, an Equivalent to a Warmer Climate.

Lastly, I shall urge but this one more Reason for this Exercise, which is not taken from a Natural, but a Prudential Consideration, from the particular Humour of most People in this Distemper, who are strangely inclin'd to think themselves in no great Danger, even tho' the Distemper is far advanc'd; they don't love to be told the Truth, tho' it is ever so necessary; but an honest Physician is to them, as *Micaiah* was to *Ahab*, he never has any thing good to say of 'em; they think they are strong enough in the Main; they'll tell ye, they should be as well as ever, if their Scurvey Cough, or the weight on their Breast was but remov'd: Now the Genius of the Sick must be consider'd, and these People who have so good an Opinion of themselves, may in some Sense be indulg'd and wrought upon, to exert their Imaginary Strength in Gentle Riding, and then they may perhaps come to enjoy that which is real.

I might now proceed farther, to consider in what degree of this Distemper Riding will be beneficial, whether any thing is to be expected from it in the second and last State of it; but this would be to run out beyond my Design of Brevity; only I shall take Notice, that it is no rare thing to meet with Consumptions, without any Putrid Fever, or any Reason to believe an Ulcer in the Lungs, or perhaps so much as Tubercles, but a continual Hectick, and a precipitate Wast of Nature by the Direful Acrimony and ill Quality of the Serum, as Doctor *Benet*, in his *Theatrum Tabidorum* observes, *Pag. 109. Tabidorum languor sine pulmonum aut visceris cujuslibet corruptelâ tacitâ vi obrepens Anglis infestissimus est, & nisi primis obediverit remediis (quod rarissimè evenit) funestus.* In this Case I can't but be of Opinion, that Riding well manag'd would be serviceable, tho' undertook very late, if there is any tolerable Measure of Strength left to put it in Practice.

I must here again repeat, that when I here speak of Riding, I understand the Habit of Riding, the want of which Distinction, has made it ineffectual to many a Man; He that in this Distemper above all others rides for his Health, must be like a *Tartar*, in a manner always on Horse-back, and then from a weak Condition, he may come to the Strength of a *Tartar*. He that

would have his *Life for a Prey*, must hunt after it, and when once he finds his Enemy give way, must not leave off, but follow his Blow, till he subdue him beyond the Possibility of a Return. He that carries this Resolution with him, will I doubt not experience the Happy Effects of the good old Direction, Recipe *Caballum*; he will find that the *English* Pad is the most noble Medium, to be made use of for a Recovery from a Distemper, which we in this Nation, have but too much reason by way of Eminence to stile *English*.

OF THE *DROPSIE.*

The Second Distemper which I shall consider as subject to these Measures, is one *Species* of the *Dropsie*; that is, the *Anasarcous* Kind, from which likewise I except those, which are attended with a hard Liver, or a remarkable Obstruction of some of the *Viscera*.

This kind of *Dropsie*, thus circumstantiated, does at first View, seem not to need the Assistance of any extraordinary means to help towards a Cure, it being the most Curable of all Dropsies; and we have daily Instances of its giving way to Common Medicines, nevertheless there are such exceptions in this most Favourable Case, as give trouble enough to a Physician sometimes, and requires more than usual Application; as for Instance; sometimes a Person happens to be brought so low by an Unseasonable Purge, that afterwards Diureticks and Corroboratives will have no effect upon him, but the Case becomes deplorable, without the Rupture of any *Lympheducts* or other the like Difficulty.

Secondly, when People decline in Years, there are some extraordinary means requisite to make the Remedies exert themselves with like Success, as they do in Younger Persons.

Thirdly, in Hysterick Women it is difficult to carry off the load of Water by common means, without some such Method as I shall hereafter mention; because their Spirits are so low, that they can bear no considerable Evacuation.

Fourthly, when a Dropsie comes upon an *Asthmatick* Person, there are particular Difficulties arise, and the singular Advantages of constant and gentle Exercise in this Case are universally known.

These four different Circumstances of this Distemper, may suffice to shew that I have Colour enough for my calling in the Gymnastick Method into this Case, and 'tis the first of the Exceptions, I mean the ill Effects, which sometimes follow upon the Use of Purgatives, which have chiefly occasion'd me to inquire, whether we ought in this plain Case, thus

circumstantiated to halt between two Opinions, between Purgatives and Diureticks, without endeavouring to establish a certain Praxis upon Just Foundations.

There are none will deny, but Diureticks are the most proper and natural Remedies in this Case, if they would always succeed, because directed to the proper Emunctory, the Kidneys, and because they can go hand in hand, with the Corroborative Medicines to be given at the same time; I take it for granted therefore, that whenever Purgatives are us'd in this Case, it is because the Diureticks don't take quick enough, or in order to carry off the load of *Serum*, that the Diureticks may the sooner display their good effects, because it will be alledg'd that the *Serum* becomes so Ropy and Glutinous in the Passages and Capillary Parts, that the Diuretick cannot always act upon it. But tho' this is granted, it will not suffice to warrant the Use of the stronger Purgatives, because their manner of Acting cannot agree with this Distemper, and because those difficulties objected, may be overcome by other means.

First, the very Nature of strong Purgers, makes against this Case, it seems very preposterous, to have recourse to such Deleterious Drugs, to those *Mortis Catapultæ* (as *Ludovicus* calls the *Esula's* and such like Purgatives) in Order to the Restoring an impoverish'd Blood; if they acted only by Stimulating the Intestines, something might be said; but since it is indisputable that they pass into the Blood, and act powerfully upon it, there is no doubt to be made, but they fuze and divide it, and break its Globules, and consequently make as much Water as they carry off, which is the very Reason why Sweating is laid aside, and Salivation, tho' they both seem so proper to carry off Watery Humours; I know it may be alledg'd in Defence of these Medicaments, that the 36th and 37th *Aphorisms*, of the *Second Section* seem to imply, that a Sick Person would receive less Harm from 'em, than one that is in Health; but yet this will not excuse their Use in our Case, because tho' the Viscousness of the *Serum*, may blunt the Particles of those Drugs for a time, and hinder 'em from working so quickly, yet when once they are thoroughly imbib'd, and begin to exert their Force, they ravage the very Principles of Life, and can by no means be fit for a Person in so low a condition. But admit that the Water is carry'd off by these means, the Blood will be left as poor at least as it was before the Dropsie first appear'd; and then how can we be sure the Waters will not rise again?

Suppose an *Anasarca* follows upon an *Hæmorrhage*, which is very common, and you draw off the Water by Purging; will not the Person be just in *Statu quo*, upon supposition that the Medicines in their working did not impair Nature? but that is not to be granted, because it is impossible to suppose, that such *Drastick* Medicines, should not prey upon Nature, even while they are assisting her; and can we be assur'd that the Blood will not run into the same Colliquation it did before? Besides, may there not be some reason to suspect that the very quantity of the *Serum*, supposing it is not too Turgid indeed, may sometimes be serviceable, to the promoting the activity of the Diuretick, even as we find in the true *Ascites*, it is of some use in the Cavity of those Persons, because they often can't spare it, without certain Ruine? We don't know how much the confidence of the Fluid may conduce to the keeping its homogeneous Particles combin'd, and we ought to be very tender of doing any thing, that might tend to dissolve the *Crassamentum*, the Globules, which are as it were the very *Semen Sanguinis* (if I may so speak); for how far Nature would endure such measures, before the Sanguification would be totally subverted, would require a Dissertation, longer than my Scope will permit; but that this is sometimes done is not improbable, and I take this to be the Case of a Young Fellow I knew, who falling into a slight Dropsie, goes to an *Empirick* somewhere about *White-Chappel*, from whom he had a Dose of Pills, which gave him about 30 Stools, which sunk him so much that his Nails turn'd black, and he died in two or three days time; Here 'tis very likely the Signification was entirely extinct, and the Blood chang'd into a Preternatural Fluid, and all by the great Power of these Deleterious Drugs; and tho' 'tis likely the Quack did not know the proper Dose of his Medicines, yet one would think, this was no more than what might be expected from Ten Grains of *Elaterium*, which yet has been allow'd by an Eminent Writer.

Besides the weak State of the Blood, the Ventricle is always more or less impair'd in this Distemper, and consequently unable to be put to bear the violent *Stimuli* of the stronger Purgers, without Danger of having its Tone irrecoverably ruin'd.

It may likewise be Prudent to forbear Purging in this Case, left happily there should be some greater Obstruction in the Liver, than we are aware on, for then it might be follow'd with ill Consequences; 'tis true, if that

Bowel is really *Schirrous*, it may be discern'd, or a great Tendency toward it, will shew it self sometimes in the Greeness and Virulency of the Bile mixt in the Excrements, together with other Indications; but a slight disorder there, is not always regarded, and Brick-colour'd turbid Urines are so common in all kind of Dropsies, that we may not discern that the Blood does abound too much with Bile, and so a Purge given at such a time may do a great deal of Mischief, for the Bile is of a light Nature in Comparison of the *Phlegm*, and moves easily, and no Man knows what he does when he rouzes it; I knew an ill accident happen once upon a Purge, given by a very Eminent Physician, to a Gentleman in a *Jaundice*, which put him into the most extravagant and fatal *Hypercatharsis*: thus bold Administrations to such weak Subjects, may be attended with Tragical Accidents, but the milder and gradual measures may succeed, without such dangerous Risks, if we consider what have been the Difficulties which have lay in the way, and hindred the Operation of our Diureticks.

The ill Success of our Diuretick Method in this Distemper, is very much owing to our giving those Medicines in so small a Quantity, and to our not changing 'em for some of a quite different Nature, when one sort us'd pertinaciously does not take; that the quantity must be encreas'd, there needs no better Argument, than what is brought for the use of Purgers; for if the Blood can dispense with the Particles of a Purgative, it will certainly bear a great quantity of those which are Diuretick only. What Wonders has that Golden Remedy of *Pythagoras* done, the *Acetum Scylliticum*, when given to a proper quantity? And what may not be expected from the *Sal Succini*, which may be given to a Dose large enough to irritate the Fibres of the Stomach, and in some measure supply the place of a gentle Purger; but when it is come into the Blood it may prove Cordial as well as inciding? And now I am speaking of augmenting the Quantity of our Diureticks, I can here affirm a very strange Effect that follow'd upon an excessive Dose of *Millepedes* in an odd kind of a Rheumatick Case, for the Cure of which, several things had been try'd in Vain, by very good Advice; the *Millepedes* were given to a quantity scarce credible, to several Ounces, and gave a Relief in a little time that exceeded all expectation. This with other instances something of the like nature, every where to be met with, may convince, us that we ought to advance the quantity of these Medicines, to which if we apply the Use of Exercise, the highest Advantages may be expected: For to grant as much as the favourers of the Purging Method can

demand, that by reason of the foremention'd Ropiness of the *Serum*, the Diureticks and Chalybeates will but distend the parts, and make the Juices grow Turgid. Is there no way to remove the Dam, but by shaking all Nature at the same time? Must we blow up the House to get the Enemy out? To what purpose do we talk so much of the Animal Oeconomy, if we reduce its Rules to Practice no more than we do? We are taught the Benefit arising from the Constriction of the Muscles upon the Vessels; and can there be any Case which does more apparently call for it than this? When it is hazardous to attempt by inward Violence to dislodge the Viscous Concretions, certainly it is high time to do it by Muscular Force. This *Hippocrates* seems to be experimentally convinc'd of, by his frequent inculcating the Use of Exercises in this Distemper, Δεῖ ταλαπωρέειν *you must labour*, is his constant Expression, whenever he speaks of the Dropsie; which, whoever considers the Conciseness that is in all the Writings of that Great Man, will be apt to imagine that it carries its Weight with it, and implies the absolute necessity of acting upon the Lentor of the Phlegm, by the playing of the Muscles. Besides Exercise will help to restore the Tone of the Parts, which is sometimes spoil'd by too great a Distension, even so much as to be in a manner benum'd, which *Helmont* seems to lay much stress on, when he, according to his odd fantastick way, calls it the *Anger of the Archæus*, that won't let the Waters pass; and if there is this kind of Spasmodick Affect in the Parts leading to the Kidneys, then certainly there is as much Reason for one in a Dropsie to get into a Coach upon his taking his Medicines, that the frequent jolting may assist their Operation, as there is for one in a Fit of the Gravel so to do. The Heat that is acquir'd by the Motion of the Body, must needs comfort the Parts, and rarifie a great deal of the Moisture, so that it may the more easily pass the Membranes, as they are dilated by Exercise; and if we can by squeezing, make Water pass through Leather, the whole Skin dry'd and prepar'd, may it not much more easily pass the Membranes of a living Animal, when work'd and stretch'd by Motion, and assisted by the Warmth which that Motion produces? These may be thought little things by some, but they will be found to be of great Consequence; by such minute Measures, Nature can produce great Effects; and by a Neglect of these things, many a great Life has been lost, in Dependence upon something of a greater Name, that has had no Relation to the Genuine proceedings of Nature.

These are some of the Reasons which have convinc'd me of the Preference of the Diuretick Course, and which I think can't be overthrown, by all the Examples of the Success of Purgers, because if we compute the Ill Effects of 'em likewise, and set 'em to balance the good, the very Cures done by 'em, will seem but as so many *Splendida Peccata*. We ought not hastily to quit safe Means for those which are dangerous, only because they are a little more expeditious; when a Case is within our Reach, we ought to Establish our Prognosticks upon sure ground, tho' they may not be so quick as could be wish'd; we have other Dropsies that are dubious enough, but in this Case we ought to study to bring things to a certainty as much as possible; which how can we do unless our Methods are Uniform? It behoves the Patrons of Purgatives to assign some certain Rule, to render the Use of 'em alwayes safe, which seems impossible to be done; and it behoves those who are for insisting on Diureticks, to find out some such Measures, as may make these milder Medicines always Efficacious; which is what I have been attempting to do; and which, if I don't flatter my self, I think I have made to appear plain and obvious; for if we can't arrive at some comfortable certainty in this Case, I don't know in what we can do so; for we are so happy as to have those things as will certainly act upon such a Crasis of the Blood, as will revive and enrich it, when decay'd, tho' not always in the like space of time; and when they act too slowly, we can enforce their Virtue, by these ways I have been speaking of.

These things are no Figment of mine, they have been the Practice of Ancient Times, and are so natural a Result from a due Consideration of the Animal Oeconomy, that I cannot enough wonder that in so many Discourses upon those Fundamental Rules, there has been so little Notice taken of the Effects of the Motion of the whole individual, as superinduc'd to the internal Motions, that make up the Oeconomy; for if this had been duly regarded, it could not but have been reduc'd to Practice, and apply'd particularly to the Cure of this Distemper.

Lastly, I know these are hard Sayings to some People, who send for a Physician, as for one that deals in Charms, and can remove all their Afflictions, while they are wholly Passive; and they would take it very ill that they should be compell'd to a sort of Labour, while they carry about 'em a Load in their Limbs; but yet for all this, Nature will be Nature still; and if this be her Voice it must be obey'd. He that is in a Dropsie ought to

be Alarm'd, and look upon himself as in something the like Case with those Criminals whom the *Dutch*, upon their refusing to Work, confine to a Cellar, and let the Water in upon 'em, that they may be in a Necessity either of Pumping or Drowning. And I believe there are but few, but who, upon their being convinc'd of the real and surprising Benefit of these Means, would readily undergo the Fatigue of 'em; and things may be so manag'd, that Exercise may not be so troublesome as the Sick imagine; an easie Pad will quickly grow familiar; and where the Legs happen to be so very much distended, that there may be some danger, lest the Skin should be rub'd off, a Chaise may serve the turn.

OF THE
Hypochondriacal
DISTEMPER.

The third and last Case, which I shall expressly consider, is the *Hysterick* or *Hypochondriacal* Case; in the Cure of which the several Exercises, which I shall hereafter Recommend, may all be us'd. This Distemper falls the most under a Gymnastick Method, because the least proper to be treated with much Internal Physick; this is a Distemper which will not drive, as we say, but if kindly treated will lead, that is, will not be expell'd by Purging, Bleeding, Sweating or the like, but must be treated by more gentle and leisurely Methods; 'tis a Distemper of the Spirits, and the Vessels which immediately convey 'em; and therefore those means by which they are more immediately affected, are the most likely to prove beneficial. Here it is, if ever, strictly true, that a little Matter gives the turn, but then that little matter must be equally apply'd; we must give an equal lift to all the Parts of the Oeconomy at the same time, we must not apply to the Fluids, and neglect the Solids. 'Tis the want of this Distinction, which I take to be the Ground of all our mistakes in the Cure of this Distemper; we cure but half the Man, When I meet with a Languid Hysterick Pulse, I can easily raise it, and give a full Beat to the Artery, by Anti-Hysterick Medicines; but then what becomes of the Nerves, they are not much help'd by this, But sometimes impair'd by it? but then let the same Person have Recourse to some moderate Exercise, his Pulse shall rife as high as upon the use of Internals, but with this Difference, that the Nerves as well as the Blood partake of the Benefit. For we may distinguish between this natural advance of the Bodily heat, which is procur'd by Exercises, and that which is acquir'd by Medicines, just as we may between the Effects of the Kindly Heat of the Sun, and those of an Artificial Fire: Now in the matter of the Vegetation of Plants, and the Management of some sorts of nicer Workmanship, tho' the greatest Care and Industry be us'd to raise a gentle heat, which to our Senses and even to the Measure of the Thermometer, may seem equal to that of the Sun, yet it shall never be able to produce the

same exquisite effects, as the heat of the Sun does. And so we see in this Case the mildest and seemingly most agreeable Gumms prove Purgers to some of these People, others again can't bear *Castor*, without some troublesome inconveniences; and how much soever some People may be Rapt up with their *Sal Volatile*, and such like Preparations, I can perhaps give an instance of more wonderful Relief given in this Case, by a more Common Cordial, than ever those splendid Medicines could produce; it may not be amiss to relate it in this place, because it serves to illustrate my Design in shewing that nothing that has the least seeming Violence in it, or rather that nothing, which is not very mild and agreeable to Nature, can be of very great moment in the Cure of this Distemper. The Instance then I mean, was communicated to me by an Eminent Physician, and very Learned Writer, and is this; He was call'd to see a Maid which had been severely Tormented with Hysterick Fits for several days, and had taken plenty of the Remedies usual in that Case, without any effect; upon which he was resolv'd to try, what a good large Dose of a true generous Wine would do, considering she was a Servant, and consequently could not be suppos'd to be accusom'd to that Liquor, which would have render'd his attempt fruitless; he therefore prescrib'd some Pouders of no Efficacy, to obviate the Phancy of the By-standers, and order'd the Apothecary to ply her with some Wine of his own procuring, that he could depend upon, till she had taken a quantity, which to her might be reckon'd very large; this succeeded like a Charm, after a good Sleep, she was freed of all her terrible Symptoms the next Morning, tho' before she could scarce stir her head from the Pillow, but she fell into a Fit. And I have twice had the Occasion, to see something of the like nature my self; the first was, where a large Dose of Wine took off some very ill Symptoms, occasion'd by strong Purgers, erroneously repeated in a certain Nervous Case. But the Person had not been us'd to drink Wine, otherwise it could not have produc'd such a happy effect. I instance in these things only to shew, that the Remedies which are most proper and adequate to this Case, must be such as have something of an inimitable Mediocrity in 'em; and that Exercises do produce Alterations in the Body, which resemble the effects of such a singular and Noble Mean, is not improbable, in regard they act so equally (as I observ'd before) both upon the Solids and Fluids. And one would think the Ill Success of any thing, but like Violence, should lead us to some such Measures as these. One would be apt to think that when a Distemper, which carries as little, or

may be, the least danger of Life in it, of any whatsoever, tho' so very troublesome, when this nevertheless becomes one of the most difficult to be perfectly rooted out, one would think, I say, that this odd Circumstance, so like to Contradiction, should prompt us to look out for the real Reason of it.

Upon these Considerations I can't but admire, that the same Administrations, or with very little difference, (excepting the Chalybeates which may be allow'd in both Cases) are thought proper for Temperate Women, and Men of Intemperance, when they happen to fall into the Hypochondriacal Affect, as is frequently enough known; one would think that when the Disorder in these latter is owing to the excess of a Liquor, both wholsom enough and Cordial enough in it self, which by its too frequent use has relax'd the Nerves, and consequently impair'd the Spirits, there should be little likelihood it should be remov'd, and taken off by hot Medicines in a Solid form, which perhaps don't differ so much as most People imagine, in their real intrinsick Energy from that Noble Liquid, to which these Gentlemen owe their Malady; I say, one would think that some such surmises as these, should naturally lead us to an immediate attempt, upon the parts affected, *viz.* the Nerves, which must be done by means suitable to 'em, that is by Exercise.

Wherever there is a Dejection of the Mind, and a Propensity to Phantastick and Imaginary Fears, there is reason to suspect the Solids, that is, the Nerves are more in fault than we think for; we may consider that when a Man is Drunk, he seldom loses his intellectual Faculties to any great degree, till the Nerves are quite stress'd with the Load of Wine, and his Feet go commonly before his Reason; and if this were a proper place, perhaps, I could shew some Reasons for us to suspect the same, in the Deliriums of People in a Fever, that the intense heat must first evidently impair the whole body of the Nerves, before the Understanding will be quite lost. We don't know what a great deal of Rotation and irregular Agitation the Spirits strictly taken will bear, without any Damage receiv'd; but when the Nerves, the Container of those Spirits, are considerably affected, the Spirits contain'd must partake of the Mischief. We know but little of that inconceivable connexion of Soul and Body, but the wonderful Bond of Union, seems to terminate very much in the Fibre. For we may observe, that those Poisonous Vegetables which intoxicate, and attack the Rational Faculty, do chiefly display their Power on the Nerves, 'tis in their very

Nature, and in the least quantity to hurt the Nerves; and when Wine, tho' in its Quality most agreeable, is by the Quantity and Repetition of it, made to be prejudicial to the Nervous System, I can't imagine, how other Cordial Medicines, which must still in some Measure keep up the Stress upon the weakned Nerves can be the adequate Remedy of this Disease, but that the Nerves, must be assisted after their own way, after a manner suitable to their Nature, that is by Exercises; for it is, and ever will be one of the Properties of a Fibre, to be the stronger for Motion, the better for wearing; and it is but a Law of Nature, arising from the Necessity of the Constitution, that while the Fluids are continually wasting and running off the faster, for the Motion of the whole Individual, the other part of the Constitution, the Solids, the Fibres should by the same Means reap some Advantages proper, and in some measure sufficient to Balance the Consequences of such a *Dispendium*.

All this receives certain weight from the Argument which is so Naturally suggested to us, by the familiar and daily Observations, which every one cannot but make on the Health of the Poorer sort of People, especially their immunity from this Distemper. That it is matter of wonder that the Spasms, the Tremors, the Shiverings, the Watchings, and all the very numerous Plagues of an Hysterick Person, should not be able to rouse People into a Quest of Health, upon Measures suitable to the Causes of things; that such Painful experience should not animate 'em, into a Resolution to exchange the Pains of a sedentary, for the Indolence at least, not to say, the Pleasures of an Active Life. I am confident no one could forbear making these Inferences, and reducing 'em to practice, who has been any considerable time infested with this Distemper, were it not for the present Comfort and false Hope, which are conceiv'd from some Palliative Remedies, in too much use in this Case; I mean Vinous Spirits, and Compositions Distill'd upon 'em, which because in the beginning of this Distemper, they are found to be comfortable and really useful, entice People to have recourse to the use of 'em, oftner than they ought to do, and in time seduce 'em so much by that Delusive flash of Ease, which they give in the first Moments of their Drinking, that they cannot have a due Regard to the Evil Consequences of such a Practice; these Liquors prove a meet Charm, they creep into the Understanding, and teach People to impose upon themselves, and fansie Excuses for the use of 'em, till they come to be so blinded as to think that Health it self, is scarce an equivalent for the Pleasures which must be

deny'd, in the Abstinence from 'em. They who have brought themselves to such a Custom, are not unlike some of our Debtors, who after they have been some time in a Prison, and learn'd the way, of living an Idle Life upon other Folks Cost, tho' under Confinement, they lose all sense of Liberty, and never desire to subsist again, upon the severe Conditions of Industry and Labour; and so those who have learn'd to sip of this Spirituous *Lethe*, quite forget the value of Health and Strength; they can drown their Vapours, blunt their Pains, and rub on without great danger a good while, and therefore as for brisk Exercise, the Cold Bath, and the like, they desire to be excus'd; there's too much Danger in the Practice, there's a *Lyon in the Way*; and thus a sickly complaining Life they lead, because they will not take Courage, to use the Just endeavours after a real State of Health. These I take to be some of the Reasons, why this Distemper is so seldom totally extirpated, and is become the *Opprobrium* both of the Patient and Physician; for else it would be impossible, that People should generally resist the Consequence of that Observation which I hinted before: For if the Labour of the Poor generally secures 'em from this Distemper, and if this Distemper, whenever it seizes, is of so nice and tender a Nature, that it will scarce allow of any of the common Methods us'd in the Cure of other Distempers, certainly it behoves the Persons so griev'd, to try whether those Means which are Preservative to others, may not prove Curative to them; which, by reason that the Subject of the Distemper, viz. the Spirits and Nerves are primarily affected by Exercise, proves highly probable. For why there should be such dependance on Internals universally, and even in this Case, I can't see; to me it seems almost as Ridiculous, as if a Workman should use but one sort of Tool in working on Wood, Stone, Brass, and all other Materials.

I need not here take Notice of any of the particular Symptoms of this variable Disease, they all being liable to the same Regimen; I have already in another place hinted what Relief may be procur'd by this Method in the Hysterick Colick, and in the obstinate Watchings, which will scarce submit to the use of *Laudanum*, or at least without ill Consequences. Therefore I shall conclude they all fall under the Power of a Resolute COURSE of EXERCISE.

The Exercises most proper here are *Riding*, and the use of the *Cold Bath*; the first prepares for the second; which Rule if some People, who are the

most weakly, wou'd observe, they might secure themselves, from some of those few Accidents that have befel the too rash entrance into the Cold Bath. Instead of Riding on horse-back, Women may take a Chaise, which will allow of swift Motion, and comes little short of the Horse for Agitation of the Body; tho' I can't see any breach of *Decorum*, if a Lady, attended with a Servant, should ride on Horse-back daily for Health, if she like it best; as for those, who upon the Account of their being very Fat, have some Reason to be cautious, how they go into the Cold Bath, lest some Apoplectick Symptoms should ensue, they may have recourse to another of those Exercises, which I shall treat of, viz. the use of the *Brush*, or *Chafing*, which if us'd in good earnest, will not prove so trivial as perhaps some People imagine it to be.

These are the Chief Exercises which I would recommend in this Case, and which if us'd with Prudence and Application, I doubt not are able to effect a compleat and Eradicative Cure of this Distemper, as certainly as more violent internal Means are expected to succeed in any other Case, the Spirits and Nerves being not so much more untractable, than the Blood and Humours, if treated after a manner suitable to their Nature.

What I have said of these three Cases, may serve to illustrate the necessity of this Method in some few other Cases, which I need only Name; as the *Scorbutick Rheumatism*, which being a Nervous Case, will admit of the three Exercises I have mention'd, which if strenuously put in Practice at proper Seasons, will do Wonders in the removing of those Pains. The *Nervous Atrophy* is another Case, which may be remov'd by a Gymnastick Method, when all the Pompous Internal Medicines will not avail.

Lastly, there is one more Case, to which Riding seems to be in a peculiar manner appropriated, and that is, that Decay of Nature which is occasion'd by Passions of the Mind, which we commonly call, Breaking the Heart; here the Spirits are broke, and ruined by the stress of Thought, the Mind drinks up the Vital Fluids, and the Ravage proceeds so fast, that nothing can avail, but what can in some Measure interrupt the Eagerness of Thought, and repair in Proportion to the wast of the Spirits, which Riding seems most likely to do, because it gives an Alacrity beyond that of Wine; because the Briskness of the Motion, must take a Man off from close thinking, and such Exercise continued long, even to some Journeys, must by Tiring incline to Rest, and break off those voluntary Wakings and anxious Thoughts, which

are so pernicious; and if some intervals of Ease can be gain'd in this Case, there is hopes that Reason or Religion may take place, and the Passions may be laid: For 'tis the first Fury that is the most Dangerous and Violent; if that can be manag'd, the Point is gain'd, and there is nothing like Hurrying the Body, to divert the Hurry of the Mind.

These are most of, if not all, the Cases which fall under the Power of Exercise as Curative; as for the Benefit which may be obtain'd by Exercise, in the Gravel, the Gout, and the like, it is purely Palliative, and therefore out of the Scope of my intention in this Treatise: I shall now proceed to consider briefly, those several Sorts of Exercise, which seem proper to my Design.

And here I shall not insist upon the various Exercises of the Ancients, or all those in Use now in our Days, but shall make choice of but a few, that seem most Compatible with the Weakness and Infirmities of Sick People, and the particular Circumstances of those Distempers which I have already mention'd, and I shall begin with the chief of 'em, which is that of RIDING.

OF THE EXERCISE OF RIDING.

Upon several Accounts, this may be esteem'd the best and Noblest of all *Exercises* for a Sick Person; whether we consider it with Respect to the Body or the Mind; if we Enquire after what manner it affects the Body, we shall find that it is a kind of mixt Exercise, partly *Active* and partly *Passive*; the lower parts of the Body, being in some measure employ'd, while the upper parts are almost wholly Remiss or Relax'd; nay, where a Man is easie, is sure of his Horse, and rides loose, there is very little Action on his Part, but he may give himself to be as careless almost as if he were Seated on a Moving Chair; so that he may be said to be Exercis'd rather than to Exercise himself; which makes the Case widely different from almost all other sorts of Exercise, as Walking, Running, Stooping, or the like; all which require some Labour, and consequently more Strength for their Performance; in all which, the Muscular Parts must be put to some Stress, and some of the Secretory Vessels made to throw off too much, while others throw off too little; whereas in Riding, the Parts being incomparably more relax'd, there is a better Disposition towards an equal Secretion of the Morbisick Particles, and a less Expence of the Animal Spirits, the chief Agents in all regular Secretions; so that a Sick Person may by this means be greatly reliev'd and not tir'd, whereas by other more violent ones, it is possible he may be tir'd and not reliev'd.

As for the Parts which are more immediately acted upon by this Exercise; it is very plain they are the whole Contents of the Lower Belly, so that the Glands of the *Mesentery* and the Intestines, so frequently accus'd of Obstructions, may in a special manner be clear'd, and their Tone recovered by such repeated Agitation; which is a thing so manifest and allow'd, that it would be needless to multiply Words in the explaining of it. But there is another sort of Assistance communicated to the Intestines, which is not so

much heeded, and that is the great Alteration, which is made by this Agitation, in some of the Morbifick Particles, as they come to be squeez'd out of their several Glands into the Intestines, which in the time of Riding is doubtless in a much greater quantity than at other times. These Particles must not be suppos'd to be barely carryed off as Excrementitious, but to undergo a Change in their Texture, to be several times in a manner Cohobated, from Acid and Acrimonious, to be Volatiliz'd, and in some measure render'd inflammable; that there is some such Alteration made in the more liquid part of the Contents of the Intestines, before they come to grow hard in a true State of Health is easie to prove, and I believe agreed on by most Enquirers into the Oeconomy of Nature, and that there is some Defect in these Operations of the Bowels, in some sick People, is evident from the Consistence, Smell, and other Qualities of these Contents, different from what they are found in a State of Health; and that this Defect may be remov'd by this Exercise, seems not improbable, if we consider how immediately Riding affects those parts, that it acts as a Topick, by those infinite Succussions coming close upon one another, which must needs cause a greater Heat than ordinarily, and a better Mixture of some of the Similar Particles, and a Rarefaction of others, which after they are thus differently Modifi'd and alter'd, are many of 'em as it were chaf'd in again by that continual Agitation, and the Steam of their inflammable Parts is of Use, to keep Nature even under the Exercise; that there is something like this to be observ'd in the actions of the Bowels might be confirm'd, by what Glysters are known to do. I would not willingly verge towards the Fraud of an Hypothesis; I may be allow'd to have had some more than Common Occasion, to put me upon making these Observations, having some time ago been so unhappy as to labour under as severe a Flux, as perhaps ever was known, which held me about a Year and a Half, attended with Vomitings, and most unsupportable Nervous Symptoms; during all which time nothing reliev'd me, in the greatest Paroxisms of it, like gentle Riding, in so much that at last I was forc'd to be in a manner always on Horseback, to have the Pressure on my Bowels rebated, and my Spirits a little refresh'd. The Comfort which I found by that means, I think must be attributed to some such *Phænomena* as I have above mention'd; for tho' I will grant, that Riding was more beneficial to me under those Circumstances, than it would be to another, because of those Nervous Symptoms; yet how Particles so exquisitely Pungent and Acrimonious, should be retain'd and blunted and

made useful, as appears from the Evil Consequences of too many Evacuations; how this should come about, but after such a manner as I have above hinted, I cannot understand; 'tis easie for those who think in hast and superficially, to be deceiv'd with the first appearance of things; but when once Men are calm enough, or under a Necessity to think closer, they are more likely to come to the Truth of such *Phænomena* as these; and to those who do allow themselves to deliberate before they are Positive, I doubt not but what I have asserted, will appear reasonable; and perhaps I should not be so much out of the way, if I should add, that some of the *Stercoraceous* parts of the Contents of the Intestines, are not in a strict Sence to be reckon'd Excrementitious or useless, since tho' I don't believe Digestion is perform'd by Putrefaction, yet I believe Putrefaction is a great Medium for the opening of Bodies, and the extracting inflammable Parts out of 'em; as we see a little Greenish *Hay*, when it comes to be Putrefi'd, shall become inflammable; and there being inflammable Particles in the Intestines, 'tis probable they may owe their Origine to some such Cause, and not to the first Chylification in the Ventricle. I would not be thought to bring these Reasons, as if I believ'd Riding would Cure a Flux, I don't believe any such thing, unless upon some very singular Circumstances, and therefore I have not plac'd it among those Distempers, which appear to be Curable by Exercise; but I only draw this Consequence from the Palliative Relief, which Riding will afford in the time of a long Flux, that some pernicious and disagreeable Particles, may receive such an Alteration while in the Intestines, as to become fit to be re-absorb'd by the several Vessels of those parts, and convey'd with great Advantage into the Blood again, which is making things to go on in a Round towards a Cure; Nature her self doing the Work, without forcible Evacuations, which tho' never so gentle in some fine Constitutions, can scarce be born, and without much Physick, the very Morbifick Matter being so alter'd and dispos'd in one part of the Body, as to be useful in another. I have insisted the longer on this Point, that I might make it as plain as possible, because I think it is of so great Moment in some Distempers and some Constitutions.

What relates to the Breast, I have had occasion to Explain before; and for the Head, tho' I can't say it is immediately affected by this Exercise as the Lower Belly is, yet there is one Benefit accrues to it from Riding, which by reason of the Disuse of Exercise in Cases of Sickness, is not taken Notice of, and it is this, the great inclination to Sleep, which a Sick Man finds if he

lies down on his Bed as soon as he comes off his Horse; for as the Motion of a Coach does more or less dispose all People to Sleep, and the swifter it goes, the more we are inclin'd to Doze; So the Motion of a Horse being swifter, and the Posture relax'd as to the Head and upper Parts, tho' a Man does not perceive any thing of such an inclination, while he is Riding and upon his Guard, without any thing to lean on, yet there is so much of the Impression of that Motion remains upon him, for sometime after he lights off his Horse, that if he throws himself presently upon his Bed, especially if he drinks some small Draught of wholsom Ale or Wine first, he will quickly be in a Sleep, which upon several Accounts must then be very Beneficial; this is a Truth so certain and so valuable to distressed, infirm People, whose Nights are often more troublesome than the Days, that it is a wonder what should keep Men from attending to Nature, and falling into such just Measures that Art it self cannot afford. What can be more applicable to all the Circumstances of Consumptive People, than after that by Moderate Riding, they have dispos'd the Humours for each Secretion, they should by such short and Refreshing Sleeps compleat those Secretions? When moreover by these Means, they may be enabled to deny themselves those latter or Morning Sleeps, in which they are so apt to run into Colliquations; I know some may please to be so witty as to call this Nursery, rather than a Management worthy of a Physician; but yet I will appeal to any that are Sober, Calm, and free from Prejudice, whether if they allow that this Exercise, does dispose to Sleep as I affirm, upon this Supposal, any thing can more exactly hit the miserable Circumstances of those Persons. To the Sick, these little things are of great Moment, and in such seemingly little things as these, the Accurate Management of the Ancients consisted, by which they were sometimes enabled to accomplish, that which we, for want of those Measures, do sometimes fall short of.

As to the other Property of this Exercise, it may be convenient for me to make some Apology, before I enter upon the Mention of it, because it is such, as cannot be well understood, but by those who are Conversant with *Sanctorius*, upon one of whose above-mention'd Maxims it does depend, viz. Upon that which shews the great Increase of the insensible Perspiration by *Pandiculation* and Gauping; now I hope the taking Notice of this, will not be thought odd in an Age, of which it is one of the Good Qualities, that Men will not take up with the old superficial Way of accounting for things by Occult Qualities, *Putredo's*, and the like, but enquire into the *Modus* of

the more Abstruse Actions of Nature, and will be convinc'd, that whatever are the legitimate Measures that she takes, they cannot be thought little or uncouth, seeing 'tis by such *Minima*, that she comes to be able to compleat her great Things. If therefore by Gauping, this Perspiration is so very much promoted, as has been discovered, and adjusted by the Experiments of that Admirable Author, we may reflect upon how little things our Deliverance from Feavers, and other Inconveniences, does depend; nothing being more common upon taking Cold, Surfeits, or the like, than for People to Gaup often, till the offensive Matter is let out, and consequently it is very apparent, that whatsoever will promote the Pandiculation must be beneficial, when the Perspiration is obstructed; and this, tho' it cannot be effected by any Internal, may be done by Riding, which will dispose all People, the Healthy as well the Sick, more or less to it. I know it may be alledg'd, that all People when they are tyr'd, are more or less apt to Gaup and Retch, but yet it cannot be said, that Thirty Miles Riding is a Tying to a Healthy Man; and yet let any one observe, if that or less will not dispose all People to this Affect, unless they over-rule it by Drinking of great Quantities of good Liquor, which I believe will not always suppress it neither; but for those who are Sickly, the least Use of this Exercise disposes 'em to this Method of Nature, which perhaps no other Exercise will do, unless they are tyr'd by it; which shews how much Riding is preferable to other Exercises for Sick People, because it does some way or other act upon the Secret Springs of Nature, after a more peculiar manner, and therefore more proper for the promoting that easie and even Evacuation.

There is another Property of Riding, that it always gives a Freshness to the Countenance of those who use it, which lasts for some time, and will appear upon but once Riding, and the weakest and most infirm Person shall discover something of this in his Cheeks after this Exercise; now I would fain know, what Noble Cordial, whether Solid or Liquid, can do thus? They may cause a Flushing, but can produce nothing of this Natural Aspect; and what can more plainly discover to us, that there is something inimitable which results from the equal and gentle Pressures of the innumerable and invisible *Vascula* of the whole Body together, and that that Action which can produce such an Appearance upon but one single Application to it, may be sufficient to display the greatest and most wholesome effects when continued on gradually, as it ought to be; and to object against the Certainty of these Measures, because they must be slow, is just as Wise as it would be

to assert, that the hand of the Dial does not move, or the Budding Leaf encrease, because we cannot discern the Motion of either of 'em. Nothing certainly could keep us from Regarding these Tendencies of Nature, but the excessive Variety of Medicines, with which we are so gloz'd, that we overlook Her gradual Progressions, either to Sickness or Health, and think to force Her in all Cases by the Power of Art; whereas in a great many Cases, she will baffle the boldest Administrators, when by gentle and suitable means she may be reduc'd, to her true State. The Famous *Cornaro's* Case, and many others might be alledg'd to shew how great Changes may be procur'd, by a strict attendance to the demands of Nature, and that it is seldom too late to aid Her in a Natural way, agreeable to her Weakness and without the Oppression as I may call it at such a time, instead of the Assistance of much Physick.

Add to all this the Vivacity, the Gayety which does alwayes more or less result from brisk Motion, whether it is caus'd by the spirits expanding themselves, or the Fibres dilating themselves to take in a greater quantity of the Spirits, it is hard to determine, and perhaps of no great consequence if we could; but that I may represent the Sense we may conceive of this, I think I have no reason to be asham'd to borrow for once more an Illustration from that Noble Beast, to which this Exercise I am treating of is owing; It is a known Case then, that if you take a Horse of the best Spirit, and of the best Keeping, provided he is not Vicious, as they call it; if you mount this Horse, and walk him or keep him to a pretty slow pace, you'll find him quiet enough, but if you once put him on to a larger Pace, he can't contain himself, but will grow troublesome, and press for a swifter Career, than perhaps his Rider would desire; which plainly shews, that there is something in the Animal Oeconomy, which *crescit eundo*, which gathers by Motion, and which can't perhaps be made to display it self so well any other way; for this must not be thought to be wholly owing to high Feeding, but to the degree of the Motion; for the same Sprightliness or Courage will appear proportionally in any sort of Motion: And but a slow Motion in some Cases does not want its good Effects; those who are Judges of the Art of War, tell us that it is not best for a Body of Men to stand still and expect the Enemy, but to keep in Motion while they are drawing to the Battle; and in the time of a Siege, they make it a Rule, to remove their Men from one Post to another; that their Spirits may be kept up by their being in a continual Diversion. We are as subject to the Impressions of Motion, as to those of

Sound and Harmony, and both are able sometimes to inspire a Flash of Courage into the Mind, that is not to be despis'd; and as one was of Use to drive away the Evil Spirit of Old, so the other may be of Service, to dispel the Hypochondriack Cloud, the desponding imaginations of Sick Persons; a Man may be able by this means to rouze himself, and shake off that *Incubus* of the Brain, that lies brooding of Causeless Fears and Doubts, to the great hindrance of all his Endeavours after Health; it is no small matter for a Person to hope and believe that he shall do well, it is some Advance towards a Cure to have so much Courage, "Ἦν φόβος καὶ θυμὸς, &c. *Si Metus & Tristitia multo tempore perseverent, Melancholicum hoc ipsum*; As *Hippocrates* observes in one of his *Aphorisms* of his *fifth Section*, Fear and Sadness are sufficient to create a Distemper, and therefore may be very well thought to obstruct greatly the Cure of one; those Passions cause the Motion of the Heart, and the Beat of the *Artery* to be weaker and consequently must proportionably lessen insensible Perspiration, which depends so much upon the Vigour of that Motion: We see a more than usual Application to Business and Intensness of Thought for but a few Days, shall cause an Alteration in the Countenance of a Healthful Man, and make him begin to look Pale and Wan; how much more then must it prejudice a Sick Man, to be always musing on his Distemper, which he can hardly well forbear neither, when he knows there is real Danger in this Case? but all this Anxiety will be very much prevented and interrupted by Riding, and a Man will naturally come to take heart and think well of his Case, when he finds he can procure such Temporary or Periodical Relief, if I may so call it, such intervals of Ease, as in the time of Riding, he is sure more or less to enjoy.

These things are so agreeable to Nature and Reason, that I am confident they can't but gain reception with those who are acquainted with this Exercise; no Man can be an Enemy to Riding, but he who is ignorant of it; and the generality of Men are by their Employments and Affairs kept so much from the Practice of it, that they for the most part judge of it by what they have experienc'd on a Journey, where an indifferent Horse, bad Ways, and other Inconveniencies, make Riding rather a Toil than a Pleasure: Whereas he who designs to make his Riding turn to account, must make it a Pleasure; he must retire to some Place, where he can have the open Field for his Range, he must find out a Horse that entirely suits his Humour, and then it will not be easie for him not to delight in a Creature which will perform all he expects from him, that takes Pleasure in what he is put upon, and

delights in his Rider; a Creature, which (considering the many other Beasts that are Serviceable for Draught or Burden) seems to be made almost only for the Defence, the Pleasure and Health of his Master; and which has so many excellent Qualities above all other Beasts, that there is no Man upon Earth, whose Gravity or Dignity is so great, as not to allow him with some Pleasure to take Notice of 'em, if the Exercise alone will not satisfie; there is Variety of the Pleasures of the Field, some of which any Man may make agreeable to his Humour; there is variety of Chace, both Violent and Moderate, a variety so great, that Providence seems to have appointed it to be subservient to this Exercise, that Men may divert themselves with Pleasures, that will keep up the Vigour of the Mind, instead of those soft Effeminate ones, which generally take place more or less, where this is laid aside; add to all this the pleasure a Man conceives when he finds his Health returning, which will make him delight in the means of his Recovery, and persue with Cheerfulness that which before perhaps seem'd indifferent to him; so that an Active Life, when a Man has laid aside his timorous Prejudices, and is let into the tast of it, will be found not only to have its Advantages, but its Charms too; and he who indulges himself long in it, will think it not a Paradox, that there should be an Active Luxury, which may exceed all the Passive Enjoyments of Sloth and Indolence. I have insisted the more on the Pleasure as well as the Benefit of this Exercise, because there are some Constitutions of so fine a Make, or else so impair'd by some Hereditary Stain, that it must be slow and gentle means that can Act upon 'em to any purpose, and the taking Pleasure in those Means must greatly contribute to the Relief they are intended to give.

Tho' what I have said, may I hope carry weight enough with if, to convince any that will give themselves leave to enquire into the Causes of things; yet because Examples have so great a sway with some I shall add a few instances of the Effects of this Exercise, and I shall first relate the History of the Cure of Dr. *Seth Ward*, then Bishop of *Salisbury*, which I have Translated from Dr. *Sydenham*.

Nostrorum quidem in Sacris Antistes, Vir Prudentia, &c. "One of our Prelates, a Man Eminent for Wisdom and Learning, after that he had for a long time given himself intemperately to his Studies, and with the whole Stress of his Mind, which in him is very great, apply'd himself too much to close Thinking; he fell at length into the Hypochondriacal Distemper, which

continuing a good while, all the Ferments of his Body were vitiated, and all the Digestions quite subverted. He had more than once gone thro' the Chalybeate Course, He had try'd almost all the Mineral Waters, with Purgings often repeated; as likewise Antiscorbuticks of all kinds, and Testaceous Powders, in order to the Sweetning of his Blood. Thus what with the Disease, and what with the Cure, continu'd for so many Years together, being just not quite destroy'd, he was seiz'd with the Colliquative *Diarrhœa*, which in the Consumption, and other Chronical Distempers, when all the Digestions are quite spoil'd, is wont to be the Forerunner of Death: When he at length consulted me, I presently consider'd, that there was no more place left for Medicines, since he had taken so many, and those so efficacious to so little purpose; I advis'd him therefore for the Reasons above-mention'd, to commit himself wholly to Riding for a Cure, beginning first with small Stages, such as were most suitable with so weak a Condition; in so much, that if he had not been of a piercing Judgment, that could discern the Reason of things, he would not have been induc'd, to try that sort of Exercise. I desir'd him to persist daily in that Practice, till in his own Opinion he was well, encreasing his Stages gradually every day, till he could come to Ride as many Miles in a Day, as the more Prudent and Moderate Travellers usually do in one day, when upon the account of their Affairs, they set out on a long Journey; that he should not be solicitous as to what he Eat or Drank, or have any regard to the Weather; but that he should like a Traveller, take up with whatsoever he met with. To be short, he set upon this Course gradually, Augmenting the Distance of his Ridings, till at length he came to ride twenty, nay thirty Miles a Day and as soon as he perceiv'd himself better after a few days tryal, he was Animated with the wonderfulness of the Event, and persever'd in the same Course for some Months; in which space of Time, he rode several Thousand Miles, as he told me himself, until he was not only well, but had acquired a strong and robust Habit of Body."

And Dr. Sydenham, tells us in the same place, that he Cur'd some of his Relations of Consumptions, by putting 'em upon Riding much, of whom he says, that it was altogether out of the Power of Medicine to help 'em. *Cum certò sciam me, vel Medicamentis quantivis pretii, aut aliâ Methodo, quæcunque demum ea fuerit, nihil magis iisdem proficere potuisse, quam si multis verbis hortatus fueram ut recte valerent.*

A Clergyman, with whom I am acquainted, living in the Country, happen'd some years ago, to fall into a lingering *Diarrhœa*, which hung upon him some Years, and eluded the force of the best Medicines of all sorts, and brought him so low, that he had no hopes of Recovery left; when he was in this Condition, a Physician of the City advis'd him to try what Riding would do, not a slight tryal or two, but a close application to it; and his Physician told me himself, that he charg'd him to keep to a brisk Motion, and gallop as much as he could, enjoyning withal a very strict Diet, that if the Disease should be check'd by the Exercise, it might not by any improper Food, have occasion to break out again. He set upon this Course in his own Grounds, which are very large and spacious, and by these means was restor'd to perfect Health again. 'Tis manifest, this Case was a Colliquative *Diarrhœa*, which at long run had sunk all the Digestions and brought Nature into a kind of Universal Gleet, so that it came to be properly and solely the Object of Exercise; whereas a New *Diarrhœa* or Dysentery, when the Humours are Turgid and Acrimonious, is solely the Object of Medicine, and so far from being to be Cur'd this way, that nothing would be more absurd than to attempt it; for 'tis the debilitated Fibres that Exercise restores, and immediately affects; and whenever Exercise makes an Alteration in the Fluids, it does so by the frequent Working and Constriction of the Fibres, which in a fresh *Diarrhœa*, before the Genuine Acrimony that occasions it is spent, would be to no purpose.

A *Northamptonshire* Gentleman, who about two Years and a half ago, came up to Town, and liv'd in *Hogsdon* Square, was taken Ill and sent for me; I found the chief thing he complain'd of was a Colick, but he had other Symptoms, which made me suspect he was beginning to be Cachectick. He was averse to much Physick, and took nothing but the *Elixir Salutis*, which gave him Ease, but he continued indispos'd; and seeing he was unwilling to take any more things, I advis'd him to ride out a little, he having a good Pad of his own breeding in the Town; he told me, if he rode at all, he would ride Forty Mile; I reply'd, I thought a much less distance would serve, and indeed as much as I was for that Exercise, I thought five or six Miles would have tyr'd him; for he was much weakned, and his Arms trembled exceedingly, when he lifted 'em up, which was caus'd purely by the Distemper, for he was not given to drink. However, after I had started that Advice, he persisted in his Design, and in two or three days set out and rode I think to *Bedford*, or thereabouts, Forty Mile in a Day, which, as he told me

afterwards, made him so stiff, that he was laid up for five or six days; but it stav'd off all those Cachectick Symptoms that appear'd before, and in about a Month he return'd well to Town, and with so Florid a Countenance, that it could be owing to nothing but that Exercise; and he continu'd so for near a Twelvemonth, when these Symptoms of an ill Habit of Body, which I clearly discern'd was begun, broke out again, and continue upon him still. This Example may suffice to shew, that the Weakness which People commonly alledge for a Reason against Riding, is no Reason at all; it being, in some Sense, their Weakness which makes it requisite.

I will here likewise mention an Instance of the good Effects of Walking, the most common and unpromising Exercise; which I had from Dr. *Baynard*. About Twenty Years agoe a certain Gentleman came from the *West-Indies* for the sake of our Hot Bath, for the Cure of a Sort of Palsie, which was occasion'd by the Dry-Gripes of that Countrey, kind of *Colica Pictonum*, which if not cur'd in time, usually terminates in a Palsie; This Gentleman got a Calash to carry him to the Bath; but it came into his Head, that he would by the way try to walk as much as he could, and when he found himself tir'd would get into his Calash; upon this Attempt he found his Limbs come to him more and more every day; and before he quite reach'd the Bath, he was perfectly well. And here it is remarkable, that *Bontius*, as great an Admirer as he was of fragrant Exoticks, in his *Medicina Indorum*, treating of a Sort of Palsie which some of the *Indians* call *Beriberii*, not much unlike to, if not the same with that I have lately mention'd, he makes it his first Rule in the Cure of that Distemper, That the Sick shouldn't give way to it, but set upon vigorous Exercise, *Sed hoc imprimis curandum est, ne (si ullo modo fieri possit) te lecto affigas decumbendo; sed vel ambulando, vel equitando, vel simili aliquo motu validiore omni conatu te exerceas.*

Dr. *Baynard* has likewise given me, in the following Letter, an Account of his Recovery from a Consumption, some Years agoe.

SIR,

*I*n Answer to your Request, concerning my Illness, as near as I can remember, I here give you in short the Matter of Fact. In the Month of October, Anno 1694, I was sent for

to my old Friend and Acquaintance, Colonel Warwick Bamfield, at Hardington in Somersetshire; I being then in London, and had been very ill all the Summer at Bath; my Case was, as I and other Physicians thought, a true and confirm'd Phthisis; for I had an habitual Heat and continual Cough, Night and Day, a very quick and frequent Pulse; I spit Blood, and exputed a viscous tough Matter, sometimes Green, Yellow, Ash-colour'd, and that in great Quantity. It would sink in Water, and smell ill and fætid when cast upon live Coals. My Flesh went off, my Stomach decay'd, and I had that Livor Genarum, as tabid People usually have, Night-Sweats, &c. so that every Body gave me over as lost and gone; but through a constant and cool Regimen in Dyet, chiefly Milk and Apples, sometimes with Honey and Sugar of Roses, and a distill'd Milk, with the temperate and cool Pectorals, together with constant Riding Night and Morning in the Air, and that on the highest Hills and Places I could find. I thank God, in two Months time my Hectic abated, Cough ceas'd, Flesh came on, and my Stomach return'd; and by continuing Riding, and other Field-Exercises, I recovered to a Miracle: And this present Year 1705, falling into the same Distemper, I was cured by the same Means, but chiefly Riding. This is very well known, and observed by all that knew me at the Bath; And I wish others, in my Case and Circumstances, may find the like happy Success. I am,

DEAR SIR,

Your humble Servant,

Edw. Baynard.

I shall here insert a Relation of a very strange Cure by Riding, which was communicated to me by Dr. Sydenham, the Son of the Eminent Writer of that Name; who was likewise pleas'd to acquaint me, That he himself took a Journey into *Scotland*, that he might get rid of a Cough, which seem'd to threaten a Consumption, and that his Journey took it off. But the Cure I am going to mention, was of a Gentleman who is related to the Dr. and now

living in *Dorsetshire*, who was brought so low by a Consumption, that there seem'd to be no Possibility of a Recovery, either by Medicine or Exercise; but it being too late for the first to do any good, all that was to be done, was to be expected from the latter, tho' the Dr. did not think that Riding would then do. However the poor Gentleman, seeing there were no other Hopes left, was resolv'd to attempt to ride into the Country; but was so extremely far gone, that at his setting out of Town, he was forc'd to be held up upon his Horse by two Porters; and when he got to *Branford* or *Hounslow*, the People of the Inn, into which he put, were unwilling to receive him, as thinking he would die there, and they should have the Trouble of a Funeral; but notwithstanding, he persisted in his Riding by small Journeys to *Exeter*, and got so much Strength by the way, that tho' one Day his Horse as he was drinking, lay down with him in the Water, and he was forc'd to ride part of the Day in that wet Condition, yet he got no Harm by it, but came to the abovemention'd place considerably recovered; where thinking he had then gain'd his Point, he neglected to ride any more for some time; but finding himself relapsing, he remember'd the Caution which Dr. *Sydenham* had given him at his setting out, That if he should be so happy as to begin to recover, he should not leave off Riding too soon, for he would infallibly relapse and die, if he did not carry on those Measures long enough; so he betook himself to his Horse again, and rode till he obtain'd a perfect Recovery.

And I have lately met with a Gentleman of this City, who upon the Advice of the same Physician, set upon a Course of Riding, and recovered of a Consumption, in which he was very far advanc'd; and had try'd a Milk-Diet, and other proper means to no purpose, and all along spit Blood very much. This Gentleman set out on a Journey to *York*, and by Riding close Day after Day for about Ten Weeks; in which space of time, he rode by Computation a Thousand Mile, he return'd healthy and well to Town.

It is to be consider'd from these two last Cases, that the Riding through Variety of Airs in a long journey, is of great Consequence to Consumptive People, and is much better than riding constantly in one Air; besides the new Scenes that appear every Day in a long Journey, create some sort of Amusement in the Minds of Sick Persons, that is not to be thought altogether contemptible.

But I have been the more willing to insert these two last Cases, because they do manifestly justify that well-grounded Distinction, or as I think, I may rather call it, Discovery of that Excellent Physician, whom I have so often cited, *viz.* That it may be too late to force any one Secretion to good purpose; and yet it may not be too late to move all the Secretions of the Body at once, equally and gently by moderate Riding; which I doubt not will be found, by all who shall try it, to be a real Truth, and of the greatest Importance, tho' it happens to be so difficult of Access to the Understandings of some People, and so cross to the Expectations of this Age, that there are Thousands of *Naaman's* Opinion to be found, who will choose to suffer any thing, rather than be convinc'd, that there can be so much Healing in the *Waters of Jordan*.

I could give several more Instances of this Nature; I could bring the Example of a Young Lady, the Heiress of a very Eminent Family, who ow'd what ease she had under a certain Distemper, chiefly to frequent Riding on Horseback, and to whom the being put out of that Method prov'd Fatal, when Her ordinary Physician being out of the way, another, who mistook her Case, took wrong Measures. But I only mention this, to shew that it may not be so incongruous a thing, and altogether without Precedent, to recommend these Measures in some pressing Circumstances, even to that tender Sex; who if they knew the surprising Advantages, that may sometimes be obtain'd by this Exercise, would I doubt not break through the Mode to come at 'em: No Woman living would bear some of the severer Hysterick Symptoms, if she knew any way to get rid of 'em; and I am widely mistaken if some of those Symptoms, do not as it were point out to us the clearest Indications for these Measures; as in those Women who have been long distress'd and broke with this Distemper, we may observe sometimes, that their Spirits are so scatter'd, or the Nerves so impair'd, that they can't well bear any thing that pleases, or displeases very much, without some disorder; if they happen to desire a Thing very earnestly, they can't wait a little while for it, without some visible uneasiness; and tho' they are sensible of this, and their Reason is as strong as ever, yet they can't command themselves, because the Animal Spirits, the *Medium* by which the Rational Soul exerts it self, are so broke and confounded. The same is likewise indicated by those intense Hysterick Shiverings, which sometimes tho' more rarely are to be met with. Now if Women, who happen to be thus Tormented, believ'd that a Recourse to this Exercise would relieve 'em, I

leave it to any one to judge, whether they would dispute the putting it in Practice.

What I have said concerning Exercise, I hope may suffice to convince any Man, that the Power of Healing is not confin'd to the Drug only, but that this course may come in for a share also, and be esteem'd upon a Level in due place with common Physick. And if I should venture to say something greater of it, I should not speak my own Fondness or Phancy, but the Opinion of one who is known to have been a very Ample Judge of the Demands of Nature, I mean Dr. Sydenham, with whose Encomium on this very Exercise, as he has given it us in his *Dissertatio Epistolaris*, and his Treatise of the Gout, I shall conclude. In the first of those abovesited Places he has these Words. *At verò nihil ex omnibus quæ mihi hactenus innotuere, adeo impensè sanguinem spiritusque fovet firmatque, ac diu multumque singulis fere diebus Equo Vehi. Cum enim in hac Gymnasticæ specie impetus fermè omnis in Ventrem inferiorem fiat, in quo Vasa Excretoria (quotquot fœculentiis, in sanguinis massa stabulantibus, educendis à naturâ instituuntur) sita sint, quæ tanta functionum perversio, aliâve Organorum Naturalis impotentia vel fingi potest, cui tot succussionum millia eodem die ingeminata, idque, sub dio, opem non attulerint? Cujus Calidum innatum usq; adeò deferbuerit, ut hoc motu non excitetur & denuo effervescat? Quæ verò sive præternaturalis substantia, sive succus depravatus in aliquo harum partium sinu recondi potest, qui non hoc Corporis Exercitio, vel in statum naturæ consentaneum perducatur, vel quaquaversum dissipetur elimineturque? Quid quod sanguis perpetuo hoc motu indefinenter agitatus ac permistus quasi renovatur ac vigescit.* And in his Treatise of the Gout, he thus expresses himself with some Exultation. *Sanè diu multumq; mecum reputavi, quod si cui innotesceret Medicamentum, quòd & celare vellet, æquè efficax in hoc Morbo (scilicet Podagrâ) ut & in Chronicis plerisque, ac est Equitatio constans & assidua, opes ille exinde amplissimas faciliè accumulare posset.*

OF CHAFING.

The next I shall recommend, is a Cutaneous Exercise; *Chafing of the Skin*, or as we usually call it, the Use of the *Flesh-Brush*. It is very strange that this Exercise, which was in such universal request among the Ancients, of which they have wrote so copiously, have given us so many Rules and Distinctions for the use of it, which they put in Practice, in almost all Distempers, and without which, scarce any Man of tolerable Circumstances pass'd a day, either in Sickness or in Health; I say, it is strange, that what was so much esteem'd by them, should be so totally neglected and slighted by us, especially when we consider that their Experience agrees so exactly with our Modern Discoveries in the Oeconomy of Nature, viz. That there is so great a disproportion between the Evacuations perform'd by the Skin insensibly, and all the others put together; that the first exceeds all the rest by many Ounces. One would be apt to think, that this Theory should convince us, that the Ancients did find their Account in those diligent Frictions, and that they really answer'd their Expectations in the several Cases, in which they made use of 'em; and that we, who live in a Colder Climate, have much more reason to expect great advantages from this Method, if we would use it to some purpose, with Continuation and close Repetition. If a Person happens to be a little more costive than ordinary, what a Concern is he in for it? What Doses of Purging Physick are repeated to take off this suppos'd Evil; which at the same time is frequently obviated by a larger *Diaphoresis*, which at such times is often sensible in the Palms of the Hands, and very often not sensible, but yet real, and to the greater Benefit of the Person, than a Laxity of the Intestines would have been. But if six or eight Ounces of the *Materia Perspirabilis* is kept in, which is of far worse Consequence, than the like Weight of the *Fæces*; no body is very solicitous about that: and if it discover it self in a Cold or Headach, presently there is Recourse to Purgatives; and if it be the Summer time, perhaps the Purging Waters are drank so long *de die in diem*, till Nature lose the way she has been accusom'd to; and perhaps never comes to be able to make the same Discharges for Quantity by insensible Perspiration, as she

did before she was thus violently forc'd out of her way. This was not the way of the Ancients, they were for stimulating and soliciting that part, which was primarily defective, that they might reduce it to an Ability to make its wonted Discharges; so that where there is a great Lett of insensible Perspiration, which in some Cases is easily discover'd by the Smoothness and Dryness of the Hands, it is certainly most natural to endeavour to stimulate the Glands of the Skin by rubbing; which by the Colour it brings into the Skin, Sufficiently shews what it is able to do, if us'd long enough. And certainly we ought to have regard to this sort of Discharge above others, because it can supply the Defect of others, better than any of the others can supply the Defect of this; and because it is perform'd by those Vessels, which are by all now allowed to be the grand Emunctory of the Body, that is, by the true Skin, and all its innumerable Glands.

These Reasons are so Natural, that I can't imagine what should have hinder'd the putting this Method in Practice in some Cases at least, unless it be, what I have somewhere observ'd before, the general Impatience of most People, who can't be brought to think well of a Method which does not surprize with some present Alteration, without considering, that if such a Method will after a time cause a good Alteration, it is worth their while to wait for it, and perhaps the best Course that the Nature of the Case will admit of. Now that the Efficacy of a *general Chafing* may be made to appear so valuable, as to encourage any one to wait for the Effects of it, let us but consider it in a particular familiar Case, which is the Cure of a *Ganglion*, a Tumour in a Tendon, occasion'd by some extravasated Juices between the Coats. This little white Swelling is commonly taken off by frequent Rubbing; and tho' no Alteration appear for a Week or two, yet if you persist longer, it certainly vanishes; which plainly shews, that if *Chafing* can produce such an Effect, in a Part which is cold, and comparatively exanguous, what may we not expect from it, when apply'd to the *Cutis*, which is so warm and succulent, and ready to give forth such copious Exhalations? This Instance, in my Opinion, does sufficiently illustrate the thing, tho' we see the good Effect of it in another Case too, viz. The *Rickets*, which is caus'd by the neglect of exercising and *Chafing* the Limbs of the Child, and which every Nurse knows may be Cur'd by so doing, if it has not been of long Continuance; or at least that those Exercises are equally prevalent with all the Internal Administrations.

If these things are so, why should not we carry on this Method to some of the other Cases too? Why should not we go to work even in a true Rheumatism, after the Inflammation is abated; to fetch out the *Mucous Gelatinous* Substance, which has been thrown put into the *Interstices* of the Muscles? This course would mould and break that Viscous Matter, and render it more fit to be absorb'd and carry'd off, or discuss'd, by dilating the Membranes, and making 'em more fit for a Transpiration; and withal would secure the Cutaneous Parts, from too great a Flaccidity, by keeping up the Spring of the small Fibres; whereas warm Fomentations, tho' they procure a wonderful Transpiration, yet they are apt to leave the part relaxed, and sodden in a manner, if they are apply'd too often; and thus it is easie to imagine, how proper this Course is in the other Nervous Scorbutick Rheumatism, and what good Effect it will produce, if us'd with Discretion, at proper Seasons; with other Exercises likewise.

There are moreover some *Atrophies* where this is like to prove of singular Use, by reason of the στεγνωσις (as *Galen* calls it, in his third Book, *de Sanitate Tuendâ*) the dryness and stiffness of the Skin, which at that time, seems to be fix'd to the part that it covers, and not to fit loose as at other times, and the Pores are obstructed, and the Skin of a different Hue, from what it is in most other Cases. Here it's easie to perceive that *Chafing* must be of wonderful consequence, the reason of which *Galen* gives us in a Chapter or two in the abovementioned Book, where he Treats expresly of this very Affect.

Lastly, it must needs be very beneficial to those Hysterical and Hypochondriacal People, who are very Fat; and upon that account, cannot use much Exercise, and have reason to be cautious, how they venture into the Cold Bath, and therefore I have not perfunctorily advanc'd this Method, where I have already treated of that Case, but upon good grounds; as these People can't well bear any of the sensible *Evacuations*, it is but necessary we should have recourse to the promotion of the insensible one; and perhaps if we knew the true cause of that Distemper, it would be found to arise in great part, from some Lett in that insensible Perspiration, and therefore it would be proper to endeavour the Removal of such an Obstruction by outward Means, because so few internal ones are agreeable; and if any one thinks the Breathing, that is caus'd by *Chafing*, too trivial to be call'd an Evacuation, they may be fully inform'd out of *Hippocrates* and

Galen, that they thought it to deserve that Name, and therefore they distinguish'd one degree of it for attenuating of gross Bodies. If therefore this can be brought to appear a sufficient Evacuation for these People, it is certain it is most conducive upon other accounts; because it raises and cherishes the Spirits at the same time. Upon this account of refreshing the Spirits, the Ancients made it a great part of their Ἀποθεραπία, that is, their Method for refreshing *Athleticks*, after their violent Exercises; and every Old Woman now among us, falls to rubbing the Limbs of any body that happens to be taken with an Hysterick Fit, for it diverts the Spirits from flowing too much to the Parts affected, and long acting upon the Extremity of those most sensible Fibres of the Skin, must needs agitate the Spirits considerably, and give some Strength likewise to those Parts that are so *Chafed*.

A great deal more might be added on this neglected Subject; but I think I have said enough to prove what I above asserted, viz. The exquisite Agreement, between the Practice of the Ancients, and our Theory of the *Cutaneous* Parts; and if this will not encourage any to hope for the like Advantage from the same Methods now in our days, nothing that I can say more will avail.

Of being Exercis'd to bear **COLD.**

The next and last *Gymnastick* Method I proceed to, is the Use of the *Cold Bath*; if any should wonder to find me rank this, among the several sorts of Exercise, they may consider, that it was ever reputed for one among the Ancients, and not without Reason; since it makes the Spirits recoil, and act with more united Vigour, upon the Subject-matter of the Disease, and so a Cure may be made by them alone, without any Medicinal Virtue, receiv'd through the Pores, as in other outward and Topical Applications; this comes up to the Notion of an Exercise, because it enables Nature to accomplish the Work of Healing her self.

I shall not attempt to account for the Advantages of this Practice, because that has been done already so copiously by Sir *John Floyer* and Dr. *Baynard*, and the World has already begun to experience the good Effects of it, and there are Examples enough, every where known, to justify the recourse to it, in the Cases to which I apply it; and 'tis to be hop'd now, that Men will begin to consider the Folly and Mischief of the too warm Regimen, which in health does often prepare 'em for Sickness, and in Sickness does often increase the Disease, and hasten Death, a Regimen which would be much more proper, for one that is to have a sudden passage into a warm Country, than for us who are to prepare our selves for the bearing of the Cold; a thing which we cannot fly from, and therefore ought to resist it; and which is not so formidable or dangerous, as for a great while has been thought. The Inhabitants of this Nation formerly went Naked, and were more Healthy than we are now; and the People of *Canada*, and all the Cold Continent behind *Newfoundland*, go much after the same manner, without any Inconvenience from it, but are rather fortifi'd against the Accidents they would be subject to, if their Pores were too much open'd and relax'd by too much warmth; and we may very well distinguish the Rational, from the Savage Part, by as thin a Habit as Decency will permit. It is a strange thing, that People should be fond of suppling their Skins, and keeping their Pores too open; as if a Man did not as really perspire, when

there is no sensible Moisture upon the Skin, tho' not so much, as when he is all bedew'd with Exhalations, which should not be sent out in so great Quantity, but upon brisk Exercise. If Men knew how much Sweat impairs the Skin, and inclines it to wrinkle, as *Sanctorius* tells us in one of his *Aphorisms*, they would be fully perswaded, that Nature can make her Discharges by finer and better ways, than those which are so perceptible, and that Flannel is scarce necessary or convenient on this side Old Age. The nervous parts of the Skin have certainly a very great Elasticity, and are capable of being strengthned by good and suitable Management, even to a Habit, as well as those of other Parts; and we see, that when the Glands of the Skin do throw out a very sensible quantity of Sweat in some particular Parts, these Parts grow accustom'd to the Air, or other Moisture, and receive little or no hinderance in their discharges from it; as we see the Palms of the Hands shall sweat copiously, notwithstanding the External Air immediately striking upon 'em; and none are more Strong and Healthy, than those who are wont to have their Feet wet without changing their Shooes and Stockings for it. The Stomach plac'd in the midst of the Body, and consequently exquisitely warm, is so adapted, as to bear large Draughts of the Coldest Liquor, without the least Damage, unless the Body has been extremely heated; and tho' its Office seems to require great and continual Heat, yet it is not obstructed in it, by the admission of Cold things, nor are its Glands benumn'd or constring'd, so as to hinder the Secretion of digesting Juices; and can we suppose the Fabrick of the Skin less perfect, and exquisite, when by its position it is to be immediately subject to the effects of the External Air? Can we think its Vessels are not endu'd with a strength sufficient to answer the Force and Weight of the incumbent Air? And its Glands of such a make, as that the Particles they strain shall be of so fine a Texture as to pass the Skin, when it seems to us to be too close to permit any transition? We make but indifferent use of a very good Theory, that *Sanctorius* has furnish'd us with, if we give way to these Thoughts, and encourage too frequently the promoting of Sensible Perspirations; which, be they in never so small a degree, are the effect of some Violence upon Nature, and consequently not to be compar'd with the other more Even and Regular Secretion.

Besides, we may argue from the Effects of too much Heat, and from the Distempers of Hot Countries, to instance but in one Disease (which when it seizes any one among us, their chiefest Care is to be secur'd from the Cold,)

and that is the Colick, which is the Epidemical Distemper of Hot Countries, not of Cold, and so common at *Surat*, that about Noon the whole Town shall smell of *Assa Fætida*, which they mix in most of their Dishes, to preserve 'em from that Tormenting Distemper, which the Heat of the Air does not exempt 'em from, but exposes 'em to it, by rarefying the Blood and Humours, and opening their Pores; by which Method I don't doubt but many a one among us has brought an Accidental Colick to be habitual; for being scar'd by the first Fit, they have endeavour'd to secure themselves from another, by these very Means which prove most likely to bring it on. If any one thinks this strange, let 'em remember what hapned when Muffs were worn universally, some Men were wont to let 'em hang upon their Bellies for the most part; and I have heard a Healthy Man complain, that upon leaving off his Muff for a day or two after such a Custom, he has been grip'd; from which any one in the World will infer, that the keeping the Part too warm, prepar'd it for the Ill Effect of the Air, and that the same may happen in any part of the whole Body; so that it is a Folly for People, in most Cases where the Lungs are not concern'd, to nurse up a Distemper, which was at first perhaps in great part owing to a tender way of living, and by continuing that Course of Life, must be rather encreas'd than perfectly rooted out. A great deal more might be urg'd, if my Scope would permit me, to induce People to believe the good effects that will follow upon the Exercising themselves in bearing *Cold*.

I shall proceed next to shew, what was the Opinion of the Antients, concerning Exercise in the Cure of several Distempers, and shall collect their Sentiments as briefly as possible; that as in a Sketch they may appear at one View.

The PRACTICE of the ANCIENTS.

We find then that much about the time of *Hippocrates*, the *Gymnastick* Method began to be introduc'd into the Art of Physick; whether it was brought up by the School of the *Cnidians*, or any other Society of Physicians, or whether *Herodicus* first joyn'd it to the Dietetick, and so brought it into Request, I shall not undertake to determine; but we find by *Hippocrates*, in his *third* Book, *de Dietâ*, *Sect. 12*. That with some sort of Glory he assumes to himself the Honour of bringing that Method to a Perfection, so as to be able to distinguish Πότερον τὸ σιτίον κρατέει τοὺς πόνους, ἢ οἱ πόνοι τὰ σιτία, ἢ μετρίως ἔχει πρὸς ἄλληλα. *Utrum cibus superat labores, aut labores cibus, aut moderatè inter se habeant*; as he expresses it. Pursuant to this, we find him in several places of his Works, recommending several Sorts of Exercises upon proper Occasions; as *first*, Friction or *Chafing*, the Effects of which he explains in his *Second* Book, *de Diæta*, *Sect. 42*. And tells us, that as in some Cases it will bring down the Bloatedness of the Solid Parts, in other Cases it will incarn and cause an Increase of Flesh, and make the part Thrive; for, says he, *Carnes Calefactæ ac siccatae alimentum in seipsas per venas trahunt, deinde augescunt*. He advises Walking, of which they had two Sorts, their round and streight Courses. He gives his Opinion of the Ἀνακίνηματα or Preparatory Exercises, which serv'd to warm and fit the Wrestlers for the more vehement ones. In some Cases he advises, the Παλὴ, or common Wrestling, and the Ἀκροκείρησις, or Wrestling by the Hands only, without coming close. The Κωρυχμαχία, or the Exercise of the *Corycus*, or the hanging Ball. The Χειρονομίη, a Sort of dextrous and regular Motion of the Hands, and upper parts of the Body, something after a Military manner. The Ἀλίνδησις, or rowling in Sand; and once we find mention'd with some Approbation the Ἥπειροι Ἴπποι, *Equi Indefiniti*, by which I suppose he means Galloping, long Courses in the open Field. These various Exercises are more amply describ'd by several Authors, and *Mercurialis* has Collected a very good account of 'em; they may seem strange to those who don't consider what great Expences the Ancients were at in Building Academies, or Places every

way convenient for these purposes; and as odd as some of these may be thought by us now, they were as commonly practis'd in those days, as Cupping is now in our *Bagnio's*. And tho' *Hippocrates* gives his Direction concerning these things, after his usual manner, in short Terms, yet 'tis plain he depended much on 'em, because he so frequently inculcates the Distinction of this or that sort of Exercise, to such and such a Distemper; and the People of those Times might find greater Benefit from those Exercises, than we do now from some of ours, which I doubt not are altogether as good; because they apply'd the Exercise to such or such a Medicine as the Physician thought fit, which gave it a greater Energy; and after its Operation had recourse to another milder Method, to take off the Heat or Disorder which might have been caus'd by the Medicine. In these Practices they were so dexterous and successful, that tho' their *Ætiology* strictly taken, was wrong, yet if a Man diligently attends to their Reasoning from Effects, he may be satisfi'd, that they were able by these Methods to do some wonderful Cures; and indeed they sometimes had such surprizing Success, that the abovemention'd *Herodicus*, an Eminent Master of Exercises, thought he could Cure all Distempers by those Means, and went Empirically to work upon the Bodies of those who put themselves under his Care, and was so extravagant, as to attempt to Cure Fevers by such Methods, for which he is ridicul'd by *Hippocrates*, in the *Sixth Book* of his *Epidemicks*, and the *Third Section*.

As for *Galen*, he follows *Hippocrates* in this, as close as in other things, and declares his Opinion of the Benefit of Exercises in several places; his *Second Book, de Sanitate Tuendâ*, is wholly upon the Use of the *Strigil*, or the Advantage of Regular *Chafing*; he has wrote a little Tract, *de Parvâ Pilâ*, wherein he recommends an Exercise, by which the Body and Mind are both at the same time affected. In his Discourse to *Thrasibulus*, which is a Dispute, whether the Preservation of the Health properly belongs to the Art of Physick, or to the Gymnastick Art, he inveighs against the *Athletick*, and other violent Practices of the *Gymnasium*, but approves of the more moderate Exercises, as subservient to the Ends of a Physician, and consequently part of that Art.

The other *Greek* Writers speak much the same thing, and the Sense of most of 'em in this matter is contracted in *Oribasius's* Collections. I shall only take Notice that they rely'd much on Exercise in the Cure of the

Dropsie, wherein we almost totally neglect it; *Porro motus, si quid aliud* (says *Trallian*, one of the latest of 'em) *Hydropicis conducit, præcipuè qui fit per Mare, Equum, & Lecticam iis autem qui viribus constant, etiam Itio est utilissima*; which is no more than what *Hippocrates* has advis'd before, in his *de internis Affection. Sect. 28*. He orders for one that has a Dropsie *ταλαιπωρεέτω περιόδοισι πολλισι δι' ἡμέρης, Laboret circuitus multos de die*. And he makes use of the same Word in his *Epidemicks*, and almost always when he speaks of the Regimen of a Dropical Person, implying that tho' it be a labour for such People to move, yet they must undergo it; and this is so much the Sense of *Hippocrates*, that *Mons. Spon* has Collected it into one of the New *Aphorisms*, which he has drawn out of his Works. *Celsus* says of this Case, *Concutiendum multâ Gestatione Corpus est*; and in another place, *Facilius in Servis quàm in Liberis tollitur, quia cum desideret famem, sitim, mille alia tædia, longamq; patientiam, promptius his succurritur, qui faciliè coguntur, quam quibus inutilis libertas est*. I have made choice of these Citations, that I may not be thought to have apply'd the Use of Exercise to the Cure of the Dropsie, without Precedent; and if the Ancients in their Practice found the good Effects of it, we have much more reason to expect greater Advantage from such Measures, since we have a Medicine we use in this Case, which seems particularly to demand it; I mean the Chalybeate, of which I have already spoke elsewhere.

But to return to my former Design; not only the *Greeks* but the *Latin* Writers also, are full of these Methods. The *Romans* rather exceeded, than came short of the *Greeks*, in the Prosecution of *Gymnastick* Courses; and *Asclepiades*, who liv'd in the Time of *Pompey* the Great, was the Man who brought them into the most Universal Request. He call'd Exercises *the common Aids of Physick*, and wrote a Treatise, *de Frictione, & Gestatione*, which is mention'd by *Celsus*, in his Chapter *de Frictione*, but the Book is lost. He carry'd these Notions so far, that he invented the *Lecti Pensiles*, or Hanging Beds, that the Sick might be rock'd to sleep; which took so much at that time, that they came afterwards to make these Beds of Silver, and they were a great part of the Luxury of that People; he had so many particular ways to make Physick agreeable, understood so well the Τὸ Βέλπον of his Profession, was so exquisite in the invention of Exercises to supply the Place of much Physick, that perhaps no Man in any Age ever had the happiness to obtain so general an Applause; and *Pliny* says, he by these means made himself the Delight of Mankind.

About his Time the *Roman* Physicians sent their Consumptive Patients to *Alexandria* in *Ægypt*, and with very good Success, as we find by both the *Pliny's*; this was done partly for the Change of Air, but chiefly for the Sake of the Exercise by the Motion of the Ship, and therefore *Celsus* says, *Si vera a Phthisis est, opus est longâ Navigatione*; and a little after he makes *Vehiculum & Navis* to be two of the chiefest Remedies; and I am apt to believe they were the more inclin'd to make use of the Sea-Carriage, as an Exercise, for the sake of the Vomiting, which happens at the Beginning of it, that they might thereby supply their Want of gentle Emeticks, which at times are so beneficial in that Distemper, in which scarce any of the Emeticks which they then knew, can be us'd with Safety, and it was a great part of their Industry and Sagacity to make good what they wanted in Pharmacy by other Means. As for the other more common Exercises, that were daily practis'd, as it is very manifest from *Celsus*, *Cælius Aurelianus*, *Theodorus Priscianus*, and the rest of the *Latin* Physicians. And we don't want Instances of the Cures wrought by these means. *Suetonius* tells us, that *Germanicus* was Cur'd of a *Crurum Gracilitas*, as he expresses it, I suppose he means an *Atrophy*, by Riding; and *Plutarch* in the Life of *Tully* gives us an account of his Infirmary, and that he recovered a great Measure of Health by Travelling, and excessive Diligence in Rubbing and *Chafing* his Body; and he himself in his *Brutus, seu de claris Oratoribus*, relates his Case, That he was so weak, that his Friends and Physicians advis'd him to leave off Pleading, which struck him so, that he thought he would undergo any Fatigue, rather than lose the Glory of his Profession, and so betook himself to Travelling, which with other regular Courses brought him to his Health again. *Pliny, lib. 31. Cap. 6.* tells us *Annæas Gallio*, who had been Consul, was Cured of a Consumption by a Sea Voyage; and *Galen* gives us such Accounts of the good Effects of particular Exercises, and all People so unanimously apply'd themselves with Patience and Resolution to these Practices, that it cannot be suppos'd, but they must have been able to produce great and good Effects.

If any one should ask me how it comes to pass that Riding, which I have substituted as equivalent to any or all the Exercises of the Ancients, in the Case of Sickness, was so little regarded by their Physicians, but other less valuable Exercises insisted on; it may be reply'd, that the Ancients could not recommend Riding to weak People, because of their manner of Riding; they had not the same convenience, as we have; for in those days they rode

without Stirrups, which must needs be tiresome to weak Persons. We find by comparing of Medals, that the Stirrup was an Invention of a much later date, than any of the Authors I have quoted, I think by about Seven Hundred Years; so that Riding was only an Exercise for Healthy and strong Men; besides Horses were not so common in the *Levant*, or in *Italy*, as they are with us; they were reserv'd for Military Men, or at least for Men of Plentiful Fortunes, and the Ass and the Mule serv'd for common Carriage; the Horse was a formidable Creature to People that were not accusom'd to him, and especially to weak Persons. We see such a Jolly Fellow as *Martial*, could advise his Friend *Priscus*, to have a Care how he hunted, *Lib. 12. Epig. 14.*

*Parcius utaris, moneo, rapiente Veredo
Prisce, nec in Lepores tam violentas eas.*

And in the Close of the Epigram,

*Quid te fræna juvant temeraria! Sæpius illis,
Prisce, datum est equitem rumpere, quam Leporem.*

And that this Humour still remains in those Countries, tho' we have a better way of Riding, is evident from Dr. *Baglivi*, who tells us in the *Corollaries* of the 8th Chapter of his *first* Book, *de Fib. Motrice*, That he Cur'd two Hypochondriacal Persons, *Hominem Nobilem ac alium Divitem*, a Gentleman and another rich Man, that he says were desperately Ill, by Riding on Asses in the Country Air; and I believe all will agree to think they were desperately Ill, who could despense with the Ass-trot, when their Circumstances would have afforded them the Carriage of a better Creature. The *Italians* plainly discover likewise, how little they are addicted to Horsemanship, in that Proverb of theirs which says, *That a Galloping Horse is an open Sepulchre*; and according to this Opinion, they manage their Horse-Races at *Florence*, for they make their Horses run without Riders upon 'em, something after the manner of a Paddock-Course; and to make 'em run the faster, they clap a Saddle upon 'em, cover'd with a Sort of

Tinsey Stuff, that may make a fluttering with the Motion of the Horse, and fright him that he may run the faster; and instead of Stirrups, there hangs down Straps from the Saddle, at the end of which, there are Balls full of sharp Spikes, which leap up and down, and prick the Horse as he runs. This ridiculous way of Running their Horses, shews how great is the Prevalence of that habitual Timorousness, which keeps 'em from the Enjoyment of the best and most useful sort of Riding, as the Ancients could distinguish very well, as we find by *Oribasius*, in his Chapter *de Equitatione*, who after he has said, that Riding slowly was tiresome, (which was for want of Stirrups) he goes on to tell you, *Si vehementer impellatur (viz. Equus) quamvis totum corpus laboriosè concutiat, tamen aliquid utilitatis affert, siquidem magis quam omnes aliæ Exercitationes, Corpus et præsertim Stomachum firmat, et sensuum instrumenta purgat, eaque reddit acutiora*. This is a sufficient *Encomium* of Riding, coming from the Mouth of one of the Ancients, who relied so much upon other Exercises, tho' we in these Northern Parts want no Recommendation from them, of a thing so much experienc'd by our selves; only we slight and neglect this Advantage, as we do many others, because it is common.

I have now with sufficient Brevity, consider'd how much the Ancients depended on the Assistance of Exercises in their Practice; and I leave it to any one, that is not prejudic'd, to judge whether this may not pass for one Reason, why they cur'd so well with so bad a Theory, and such indifferent Materials; when we, with our Circulation and Splendid Pharmacy, are not perhaps able to outdo 'em proportionably to our larger Acquisition of Knowledge; I say, I think this may be one Reason, for I know there is another may be alledg'd likewise, and I think I speak without any undue Aggravation.

The Power of Exercises us'd at proper Seasons, and with great and exact Patience, must needs be very great; and if it be true, that in the *Roman* Common Wealth there was no profess'd Physician for the first Five Hundred Years, there is no way to account for it but by their incredible Temperance, and Variety of Exercises; the few Chronical Distempers they had among 'em, were in all likelihood, for the most part subsequent to Acute Distempers, which no Temperance can always prevent, and the little vegetable Physick, which they could not but know, with their resolute and indefatigable Application to some of their Exercises, might suffice to help

'em to get clear of 'em. This may perhaps by some be thought rather a Gloss than an Argument, who do no duly consider their wonderful Patience in this respect; the *Pletherismus*, and *Pitylismus*, two odd uncouth Exercises mention'd by *Galen*, are sufficient to convince any Man, that they that would heartily drudge at them in the middle of the *Stadium*, would do any thing that was possible to recover their Health, and might expect the very utmost Benefit that could be obtain'd from the various Motion of the whole, or any part of the Body.

I will grant, that they carried these things too far, the whole Education of the *Athletæ* was blameable; I will grant likewise that they were too Nice in the Exercises for the Preservation of Health; I can't admire *Pliny's* Course of Life, which he gives us an Account of in the *Thirty sixth Epistle* of his *Ninth Book*. *Ubi hora quarta vel quinta (neque enim certum dimensumq; tempus) ut dies suasit, in Xystum me, vel Cryptoporticum confero, reliqua meditor & dicto, vehiculum ascendo, ibi quoque idem quòd ambulans, aut jacens, durat intensio, mutatione ipsa refecta paulum redormio, deinde ambulo, mox orationem Græcam Latinámve clarè & intentè; non tam vocis causa, quam Stomachi lego, pariter tamen & illa firmatur iterum Ambulo, ungor, exerceor, Labor.* Nor can I approve of *Spurinnas* abundant Regularity, as *Pliny* relates it in his *first Epistle* of his *third Book*, he tells us, thus *Spurinna* manag'd himself, *Ut manè lectulo contineretur, hora secunda indueretur, ambuletque millia passuum tria, mox legeret, vel colloqueretur, deinde consideret, tum vehiculum ascenderet, peractisq; ita septem millibus passuum, iterum ambulet mille, iterum resideret, vel se Cubiculo aut Stylo redderet; ubi hora balnei nunciata foret, (quæ erat byeme nona, æstate octava) in sole, si caruisset vento, ambulet nudus, dein pila moveretur vehementer, & diu; postmodùm lotus accumberet, & paulisper cibum differret.* This is a great deal too much, this is over-doing, Life is not worth such anxious Regularity, a generous Negligence is much more preferable in a State of Health; but if a Man happens to be seiz'd with a Distemper of such a Nature, that Exercise is absolutely necessary to the Removal of it, he would be wanting to himself, and very much to be blam'd, if he should think any Fatigue too great to be undergone for the sake of a Recovery; and I fansie there is scarce any Man in this Case, but would take Pains enough, provided he thought they would not be in Vain, which most People are now apt to suppose, because the World has lost the Sense of the Efficacy of Exercises, since the *Goths* over-run the Southern

Nations of *Europe*, and abolish'd and suppress'd many of their Customs, that they might the easier introduce their own; so that in this respect we are in opposite Extrems, if they prosecuted these Measures too much, we do too little; and I am very sensible how difficult a thing it is, with the best of Arguments, to influence those who are propense to the contrary of what you propose. And we are so habituated to other Methods and other Expectations, have so many Conveniences for Support and Palliation under a Sickness, that most People are supinely content rather to rub on in a Sickly Condition, that does not carry the greatest Danger with it, than resolutely endeavour by vigorous Means to be quite deliver'd from it. And if it be a Distemper that threatens immediate Peril, these Means are presently thought too trivial to have any Weight in the Cure, and People can with more Ease despair of Help, than struggle with Pain in hope of an uncertain Cure; so that in some Cases, it may be very well said of the Wise, the Rich, the Valiant, and the Mighty, as was formerly said of one of King *David's* Generals; who, because he was kill'd upon a Surprise, (which yet no Humane Precaution can always prevent,) was lamented with that severe *Epicedium*, *Died ABNER as a Fool dieth?*

THE APPENDIX.

I did not design, either when I begun or when I ended this Treatise, to add the following Account of my own Distemper, which gave me the unhappy Occasion of many of these Observations, both because it is no new thing for Nervous Cases to vary extremely in their Anomalous Symptoms, and because the Rise of the Distemper was eminently owing to my own Rashness and Folly; but having since been importun'd by some of my Friends to make it Publick, I have been the more easily prevail'd with so to do, because I conceive there scarce ever was any Instance, either among the Ancients or in our days, which does so palpably illustrate the Power of my Subject.

My Distemper was caus'd thus; I happen'd several Years ago to catch a Certain Cutaneous Infection, more troublesome than dangerous, and which I might easily have got rid of, if I had had but so much Prudence or Patience as to have submitted to the Common, tho' not very agreeable Method of Cure; but I must needs take to a more cleanly, but desperate Course, by making an Application to my Skin of a Substance well charg'd with a dangerous Mineral, and which I us'd to such a Time, and after so extreamly Rash a manner, as I believe never one did before me; I wore it for several Months by times, keeping it on till the Humour disappear'd, and then leaving it off till it broke out again; thus I did till I began to consider that this would not last always, and so I quite laid it aside; but the Mischief was begun, tho' I was not aware of it; for about a Month after I had left off that Pernicious Practice, I wak'd one Morning with an unusual Giddiness, and in a little time after, I found several Convulsions begin to come upon me, not only slight ones, in the Capillary Arteries of my Eyes and other Parts, but some deeper; as sometimes I should have one of my Hands drawn in, my Fingers brought to the Palm of my Hand involuntarily. When I found things thus, knowing what I had done, I was not a little terrifi'd, as any one may imagine, but by applying to a good Physician, by the Help of Bleeding and other proper Means, the Severity of the Symptoms went off, and I

remain'd only with a Giddiness, which was very troublesome, but notwithstanding which, I might be said to be in pretty good Health, as having a good Habit of Body, and being more strong and robust, than many People that are free from such an Inconvenience; and thus I continued for several Years without any great Alteration happening, except that once upon Drinking the Bath-Waters too long, that vertiginous Rotation was so increas'd, that I was laid up for a Month with it, being scarce able to go about without Staggering like a Drunken Man; this, considering the Strength I had notwithstanding, made it reasonable to suspect, as most that I consulted did, that there might be some Mercurial Particles lodg'd in some Excretory Ducts of my Brain, but this Extremity abated, and I remained Healthy under that Giddiness for several Years, till happening once to be more indispos'd than usually, upon the taking of a Chalybeate, the Humour struck deeper into the Nerves of the Pectoral Muscles, so that I found my Shoulders sometimes begin to be contracted involuntarily, and upon walking, I found a Shortness of Breath, which I could plainly discern, was caus'd *ab extra*, that is, by the Preternatural Pressure of those Muscles, upon my Breast, which I have sometimes resisted so far, when it has happen'd to come upon me in the Streets, before I could get to a House that I knew, that it has made my Eyes flash, and brought other Epileptick Symptoms upon me, in so much that I have been forc'd to take into a Shop, and desire leave to rest me for a time. These things continued on me a Year or more, during which time, if I had then known the Prevalency of Riding, I might have prevented a long Scene of Sickness; for declining under these Spasms of my Breast, I was at length quite confin'd to my Home, for near Three Months, all the while growing worse and worse, in Spite of the Power of any Medicines I took; when it happen'd that I was casually directed to the Use of Riding by the great Alteration I found in me, upon being one Day carried out in a Coach about a quarter of a Mile, when I was in that low Condition, that made me reflect on some of Dr. Sydenham's Notions, which, like others, I had before slighted and disregarded, and I determined to try what Riding would do; in Order to which, I remov'd to *Hamstead* for the Conveniency, where I rode at least twice a Day for the Space of Seven Weeks, without so much as intermitting one day; and I had Encouragement enough to keep close to it; for I found as Manifest a Progress in my Recovery, as ever was discern'd in any thing that is either Natural or Artificial; and I can't here omit one thing very Remarkable, and

which gave me the greatest Assurance of Success, and which shews beyond Contradiction, how peculiar this Exercise is to Cases of the Nerves; after I began to Ride, I found constantly upon the Use of that Exercise, a Tingling in the Ends of my Fingers, which was accompanied with a Sense of a gentle glowing Heat; and as I grew stronger, and persisted in those Means, I perceiv'd this Sensation proceed farther, for then I felt it in the Crown of my Head, and at length in my Tongue, in my Eyes, and at last all over my Body. This I acquainted my Physician with, who thought it a certain Pledge of my Recovery, as being assur'd that it was caus'd by the Spirits Recovering the Passage of the Nerves as in a State of Health, and so it prov'd; for as they pass'd farther and farther in the same manner, the Sense extended it self to my Wast, where I should have, after Riding, a Sense of Tingling and Heat, as strong, as if I had expos'd my Breast to the Sun, or pour'd warm Water upon it; any one may imagine what a Pleasure this must needs be to one in my Circumstances, to find my self as it were, *Renasci*, and all the parts of my Body gradually receive a kind of New Life. By this it is plain, how much I was Relax'd before, and how agreeable to the Nature of the Nerves, that most Excellent Exercise is. This *Phænomenon* has given me very Serious Reflections, when I have known some, who have stood almost in as much need of the same Measures as my self, tho' their Indispositions have Rise from different Causes, and in whom in all Likelihood the same Method of Riding would have produc'd some of the same hopeful Symptoms; I say, it has given me serious and troublesome Reflections; when upon such Occasions, I have not thought it proper, to recommend the same Means, or shew the Reasons, why I would recommend 'em, as being sure that if I did, they would be rejected with Derision. So that I have often with some Regret admir'd the Incongruity of the Circumstances of Humane Life, how little Reason Man has to be Proud, notwithstanding his Wisdom and Counsel, when even in the greatest Concerns of his Life, Custom, Mode, Phancy, and many other Circumstances, may so envelop him with Prejudice, that it may so happen, that in some Cases of Distress, one Man may be Morally certain, of what will help another, and yet it may be the most absurd thing in the World, to disclose it to him, and impossible to convince him if he should. I have the more amply related this unusual Circumstance, because it seems to me to illustrate above any thing whatsoever, the Influence of Riding, on the Nerves or Solid Parts of the Body; and as strange and unheard of a

Symptom, as it may seem to some, I can averr it is a most Solemn Truth, and what I had frequent and reiterated Occasions to observe.

After I had by these Means recover'd a good Measure of Health, I enjoy'd it about a Year and a Half, by the Help of Riding at due Intervals; and I was pretty strong, and able to make considerable Journeys, which my Affairs then required. But in those Journeys, I was so unfortunate, as to abuse that most excellent Expedient, that I had receiv'd so much Good from; for, thinking my self stronger than I really was, and Riding long Journeys beyond my Strength, I was took of a sudden, and in the midst of a Journey, with the strangest sort, of *Deliquium* that perhaps ever was heard of, in which I was forc'd to drink such quantities of Spirituous Liquors, as at another time would have made me more than ordinarily Drunk, which then could scarce keep Life in me; to these sort of Fits after some time was added a *Diarrhæa* of a peculiar and singular kind, which was attended with a Nauseousness and Inclination to Vomit, more than is usual in that Case; and when the *Diarrhæa* at any time happen'd to be a little more violent than ordinarily, a Spontaneous Vomiting ensu'd, as violent as if I had taken a strong Emetick; and what is remarkable, those Nights after my Vomiting, I should Sweat to that degree, as to make my Shirt and Sheets as wet, as if they had been dipt in Water; upon which, instead of being weakned, I was refresh'd, and freed from the greater severity of these Symptoms for some Days; which plainly shew'd, how pernicious that Psoradick Salt was which the Mercurial Particles had repell'd, and intimately mix'd in my Blood. Thus I had three Calamities upon me at once; a Flux, a Vomiting, and the *Deliquiums* I spoke of before; the last of which had something so incredible in 'em, that tho' most severely true, and known to several in this City, yet I shall forbear to relate it in this place, lest I should be thought to exceed the Truth, by telling that which scarce ever had a Parallel; I shall only say, that the Gripes and Vomiting were comparatively a Pleasure to this, I do not speak in Heat, but Seriously and Calmly, that the Vomiting, tho' so frequent and severe, was desireable, if compar'd with these most strange Faintings; which were so tormenting and insupportable, that if I had been Stab'd, or had had my Flesh cut with Knives, I am certain I could much easier have born it. Under this Complication of Misery, I was deny'd likewise the Ease, which is to be obtain'd by *Laudanum*, which I was prohibited by the Nature of my Circumstances, and by very good Advice, directed to abstain from it. And as for Emeticks, there was no using of them for other Reasons; so that

the Business was left wholly to Stypticks, which not putting the least stop to my Distemper, I resolv'd to betake my self to Riding again, having Rested so long, and would try if using it moderately and frequently, might not keep up my Spirits, as formerly it has done, and it prov'd according to my Hopes; for by diligent and fervent Application to these Means, I made a shift to support my self, and keep things in an *Æquilibrium*, under these pressing Circumstances, insomuch that I held out, even under such Severities, as I have related, no less a time than a Year and Nine Months, when at length these Evils gain'd upon me, and my Vomitings were so frequent, and incredibly violent, that they took away the Use of my Limbs, when I was forc'd to be lifted every time upon my Horse, that I might continue my Riding, tho' it were but to gain a little Periodical Ease; till that Seizure on my Limbs encreasing, I was confin'd to my Chamber, my Legs and Thighs swell'd, and in the Opinion of three Physicians, whose Assistance I then had, I was very near the End of this (as it was Literally to me a) Tedious *Journey of Life*. When it pleas'd God, that in this Extremity, one of those Physicians was so happy, as to direct me to a Mild Chymical Medicine, too seldom us'd for that intent, which stop'd my Vomiting, and created a sudden and surprising Change in me; the swelling of my Legs went off in less than three Weeks, and in about six Weeks time, from that emaciated Condition, I was as Fleshy, as in a perfect State of Health. By this it is plain, that the great stress of Vomiting threw off the Humour, and that the Swelling of my Legs was in part, if not altogether, Critical, and that the habituating my self so very much to Riding, did enable Nature to throw off the Humour that way, and support her under the Shock of those numerous Vomitings. After this, by the Use of Chalybeates, taken for a great while together, and with Riding upon it, I recovered as good an Appetite and Digestion, as I had before these Vomitings came upon me, and my Flux entirely stop'd; and for these two Years last past, I thank God I have liv'd an Easie and Happy Life.

Thus I have given a succinct and true Account, of a Long and Severe Distemper, which it has pleas'd Almighty God to lay upon me; by which it is plain, that as some Men are distinguish'd by Riches, Honours, and the like; others may be as remarkable in the Degrees of their Affliction and Anguish, and may be forc'd to pass not only Days, but Years of that which we call Life, after such a Manner, that if it were not for higher Considerations, it would be far better not to be. That in the very best of a

Man's Years, in the time of Hopes, Prospects, and Advantages, a Man may be so wretched as to be debar'd the Capacity of making use of 'em, and in a Sense, Buried Alive.

But since I have offer'd the World a Relation of my Distemper, I shall make bold to take Notice of two or three things observable in it, which may perhaps be of use to some, tho' the very same Distemper, I hope will never be the Lot of any one. And this the rather, because those Remarks are agreeable to my Subject.

First then, I think it is very clear, from the Circumstances of my Case, that where Indications arise, which run Counter to one another, which every one knows is the worst of Difficulties a Physician meets with, in such an Emergency, nothing can be so lively to balance between 'em, as the having recourse to some proper Exercise, if possible; for it is hard but it will suit with one of those Circumstances, and not improbable, that it will give help in both.

Secondly, From the Cessation of my Gripes as soon as ever I got on Horseback, I am convinc'd that Pain may be very much obviated or mitigated by the Motion of the Body, that is, by those Means a Man may make a shift to set loose from it. It was a piece of Extravagant Nonsense in the Scepticks, to pretend to Reason with Pain, and account it no Evil; a Man might as well pretend to Fence against Hunger with Resolution, and be accounted blameable for Starving through Want of Victuals; undoubtedly Pain is the greatest of Evils to the Body, and that which we cannot but endeavour to free our selves from; but till that can be done, it will be a Happiness, if we can so manage things, as to perceive less of it, or as I said before, sit loose from it; and that I think may certainly be obtain'd from the Motion of the Individual, of such or such a kind, as the Pain will best admit of. Not only Thought, but the Perception of a Pain, may be in some Measure interrupted by a swift Motion, for that Perception cannot strike so strong at such a time; a Man that should let himself to Muse on a full Gallop, would think but very incoherently, and if Thought it self may be thus broke off, certainly the Sensation of Pain, which can be no quicker than Thought, may be likewise happily interrupted and rendered more Dull. These Considerations would be of real Service to Persons afflicted with Scorbutick Pains, if they could induce 'em to a Resolute Practice of Riding; for all those Pains in the Shoulders, Back, or Hips, would gradually abate in

the time of that Exercise, as well as by the Habit of it, the Cause would be remov'd. But moreover there are other Sorts of Gestation or Voiture, that are suitable with the keenest Pains, as the Motion of a Boat, a Litter, a Chair, a Swing, and the like, in these there are no *Motûs Contrarietas*, as Dr. Sydenham calls it, no Jolting, which in some Cases, as in the Gout, is not tolerable; but they may be apply'd to the severest Cases, and I doubt not would sooth the Spirits, and give great Relief. And here I cannot but think that a good Mechanick might invent some Machine, after the manner of the *Petaurus*, or the great Swing of the Ancients, which might be of the greatest Service to People in the Extremities of the Gout, by diverting the Sensation, and turning the Edge of the Pain: A Man that has been laid up for several Weeks, would certainly find by such a Motion, as great an Alteration, as one who has been confin'd to his Bed for some time does upon his first getting up, which is very great; and certainly if the *Romans*, out of Luxury and Wantonness, could be Swung to Sleep in Hanging Beds of Silver, it may be allow'd, that a Person in Torment may use the same Means to obtain Ease. I don't know why a Motion upon a Round, should not do some Service in this Case, for there would be a sort of Sickness in the Stomach, caus'd by the Rotation, which would withdraw the Spirits from the Place of the Pain; and moreover undoubtedly dispose to Sleep. These things cannot be thought Extravagant, if they can suffice to give any the least Ease to Men under such Pains, as those of the Gout, when the *Carnificina* lasts so long, as to exceed in some Measure all Humane Patience; when *Non rectius Podagræ quam Iracundiæ Paroxysmus omnis did potest*, as Dr. Sydenham, who knew too well what it was, says of it; I say, in such Extremities, nothing ought to be thought Ridiculous, that can afford the least Ease. I could say; much more on this Subject; I could Name a very Worthy Gentleman, who not long ago had such an odd sort of a Colick, that he found nothing would relieve him in the Fits of it, so much as lying with his Head downwards; which Posture prov'd always so Advantageous to him, that he caus'd a kind of Frame to be made, to which he caus'd himself to be fastned with Bolts, and then his Servants were to turn him with his Head downwards; after which manner, he hung till his Pain went off: And I hope none will say it was unbecoming a Grave and Wise Man, to make use of such odd Means to get rid of an unsupportable Pain. But I have perhaps already said more than will be relish'd by some of those Gentlemen, who are so happy as to be really Ignorant, or to have but very slight, and imperfect Notions of the

Nature of Pain; and before they censure me, they would do well to peruse Mr. *Collier's* Excellent Essay upon Pain; where they will find such an Account of it, and such Instances of the Behaviour of Healthy and Strong Men under it, that they will the more easily admit, that for the Sick and Infirm, it may be in some Measure feasible to avoid, as it were, the Sensations of Pain, and, as much as can be, sit loose from it.

The next and last thing which I shall take Notice of, in my Distemper, is the Frequency of those Vomitings, and that notwithstanding they lasted so long on me, I receiv'd no harm from 'em. It is a great Misfortune in the business of a Physician, that he finds so many people averse to that sort of Physick, whereby he is compell'd to prosecute a Cure by more uncertain and inefficacious Means; whereas if those Persons did but consider, they might easily find Arguments to convince 'em, that there is no such Danger as they apprehend, in the use of Moderate Emeticks, where there is no *Mala Conformatio*, nothing in the make of the Breast that forbids it. They may consider that the Vomiting which comes upon going to Sea, tho' it lasts sometimes two or three days together, and all the time seems as violent as that we procure by our Emeticks, yet it never hurts any one, and does not much weaken those who have undergone it. I know two Gentlemen who were Sick all the time of their being upon the Sea, one of 'em all the time of his Passage to and from the *West Indies*, and yet they receiv'd no hurt by it. And I don't know but I my self have Vomited as much as any ten Men Living, and yet am not in the least hurt by it. And we are furnish'd with Medicines, so gentle and commendable, that they may be repeated at reasonable Intervals, without the least Danger of any ill Consequence.

From these Considerations of the great *Ευφορία* with which we bear Vomiting, tho' it seems so troublesome while it lasts; I am convinc'd that it may be accounted for after the same manner, as Gymnastick Effects are; besides that the rising of the Pulse upon the Stimulating and Irritation of the Fibres of the Stomach, and the excessive Pressure and Shock of the Glands of several parts, with the other *Phænomena* of Vomiting, shew that it does partake of the Nature of an Exercise, and it is a great Happiness for the Individual, that the Author of Nature has allotted such Secondary Uses of the Stomach, Diaphragm, and other parts imploy'd in Vomiting, that they should not only serve to through up what is disagreeable, but strengthen the whole Oeconomy in that very Act. And here if it be allow'd, that the

Irritation of the Ventricle does raise and strengthen so much, I can't but think I may venture to propose it, as worthy the Consideration of the best Judges of these Matters, whether when we use Emeticks, we ought to rest our Expectations upon a few Momentary Efforts, when we see Nature will bear the carrying on of the same Measures so much longer; that is, whether it would not be more expedient in some Cases, to give our safe and gentle Emeticks in lesser quantities, than we do, viz. so as to make the Person Sick, but not to a degree sufficient to make him throw up the Contents of his Stomach, and when that quantity of the Medicine has pass'd off after that manner like an Alterative, to repeat the same Dose, and so continue on that Sickness for several hours, without raising it to that degree, as to force the Person to Vomit above once or twice in all the time. This, I say, I think is worth Consideration, and may be of singular Advantage in some Cases of a Consumption, and in Hysterick Cases, when we do not give Vomits to cleanse the Stomach only; for by this means we can Elevate Nature very much, and procure a just *Diaphoresis*, when perhaps by the best of Cordials we may only create a Colliquation; and after this manner the Springs will be wound up more gradually; when if the Emetick passes off with strong Efforts, and very quick, after the usual manner, we give Nature only a Wrench, the effects of which are soon over. I would not be here understood, as if I would put this in Practice, I do not pretend to Authority sufficient for such Innovations, I only presume to offer these things, by way of Problem; and one of my Stature may sometimes happen to start a hint, which those who are Taller in Wisdom and Understanding may cultivate and improve to Perfection.

These are some of the Thoughts which the Severities of a most tedious Distemper have suggested to me, which if they can be of any use, to help to guide others to that which may deliver 'em from their Calamities, it would be a great Satisfaction and Recompence to me for my Pain, to perceive that it has conduc'd to another's Ease. But if these Speculations shall not be worthy to have such an Effect, I hope I shall be Pardon'd upon the Account of my good Intentions; especially seeing I am not singular, but have so very many Companions of those who have troubled the World with things of little or no Moment.

FINIS.

Printed by JOHN MATTHEWS, in
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